

Reviewed Preprint

v1 • July 8, 2025

Not revised

Reviewed Preprint

v2 • March 5, 2026

Revised by authors

An abundant merozoite surface protein of *Plasmodium falciparum* modulates susceptibility to inhibitory antibodies

Isabelle G Henshall¹, Jill Chmielewski¹, Dimuthu Angage², Ornella Romeo¹, Keng Heng Lai^{1,3}, Kaitlin R Turland^{1,3}, Nicki Badii², Michael Foley^{2,4}, Robin F Anders², James Beeson^{5,6,7}, Danny W Wilson^{1,3,5} ✉

¹Research Centre for Infectious Diseases, School of Biological Sciences, The University of Adelaide, Adelaide, Australia • ²Department of Biochemistry and Chemistry, La Trobe Institute for Molecular Sciences, La Trobe University, Melbourne, Australia • ³Institute for Photonics and Advanced Sensing (IPAS), University of Adelaide, Adelaide, Australia • ⁴AdAlta, Bundoora, Australia • ⁵Burnet Institute, Melbourne, Australia • ⁶Department of Infectious Diseases, University of Melbourne, Melbourne, Australia • ⁷School of Translational Medicine and Department of Microbiology, Monash University, Melbourne, Australia

✉ For correspondence:

danny.wilson@adelaide.edu.au

Competing interests: M.F. is the founding chief scientist and a shareholder in AdAlta Ltd., and R.F.A. is also a shareholder in AdAlta. The other authors have declared no competing interests.

Funding: See [page 23](#)

Reviewing editor: Urszula Krzych, Walter Reed Army Institute of Research, United States

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eLife Assessment

This **important** work provides a fresh perspective on merozoite surface biology and its implications for vaccine design, challenging the prevailing dogma that MSPs are indispensable invasion engines. The revised manuscript strengthens the **compelling** evidence that, although MSP2 is dispensable for parasite growth, it acts as an immune modulator of AMA1. While the study is commendable for its use of state-of-the-art technologies and the skillful application of monoclonal antibodies, the inclusion of human monoclonal antibodies and electron microscopy imaging approaches would significantly add to the importance of these observations. Overall, this work will be of considerable interest to investigators studying Plasmodium biology and vaccine development.

<https://doi.org/10.7554/eLife.107603.2.sa3>

Abstract

Malaria merozoite surface proteins (MSPs), are thought to have important roles in red blood cell (RBC) invasion and their exposure on the parasite surface makes them attractive vaccine candidates. However, their role in invasion has not been directly demonstrated and their biological functions are unknown. One of the most abundant proteins is PfMSP2, which is likely an ancestral protein that has been maintained in the *Plasmodium falciparum* lineage and is a focus of vaccine development, whose function remains unknown. Using CRISPR-Cas9 gene-editing, we removed PfMSP2 from two different *P. falciparum* lines with no impact on parasite replication or phenotype *in vitro*, demonstrating that it is not essential for RBC invasion. However, loss of PfMSP2 led to increased inhibitory potency of antibodies targeting other merozoite proteins involved in invasion, particularly PfAMA1. In a solid-phase model, increasing concentrations of PfMSP2 protein reduced binding of different antibodies against PfAMA1 in a dose dependent manner. These data suggest that PfMSP2 can modulate the susceptibility of merozoites to protective inhibitory antibodies. The results of this study change our understanding of the potential functions of PfMSP2 and establishes a new concept in malaria where a surface protein can reduce the protective efficacy of antibodies targeting a different antigen. These findings have important implications for understanding malaria immunity and informing vaccine development.

Introduction

Plasmodium falciparum malaria remains a global health challenge with over half a million deaths annually (World Health Organization, 2024 [\[1\]](#)). Worryingly, increasing parasite resistance to frontline antimalarials and mosquito insecticide resistance are decreasing the effectiveness of key control measures (World Health Organization, 2024 [\[1\]](#)). Development of the RTS,S and R21 vaccines targeting the *P. falciparum* circumsporozoite protein is a step forward for malaria vaccine development, however further improvements in vaccine efficacy and longevity are needed to protect those most at risk – children under 5 years – and enable malaria elimination (Beeson et al., 2019 [\[2\]](#); Datoo et al., 2024 [\[3\]](#); RTSS Clinical Trials Partnership, 2015 [\[4\]](#)). Understanding the function of vaccine targets remains important for prioritisation and optimisation of candidates in the development pipeline. Merozoites, the parasite stage that invades host red blood cells (RBCs), have long been targeted for vaccine development since successful blocking of merozoite invasion would prevent the repeated cycles of blood-stage parasite multiplication and associated disease. While much work has been done to understand the processes and steps required for merozoite invasion, only a few merozoite proteins have successfully had functions defined. To date, most vaccines based on merozoite antigens have demonstrated limited efficacy (Beeson et al., 2016 [\[5\]](#)) at least in part due to the extensive polymorphisms exhibited by many of these antigens.

The surface of the merozoite is covered in a fibril coat which is comprised of multiple proteins, broadly known as merozoite surface proteins (MSPs). These proteins have been proposed to mediate weak initial merozoite RBC contact (Bannister et al., 1986 [\[6\]](#); Beeson et al., 2016 [\[5\]](#); Cowman et al., 2017 [\[7\]](#); Weiss et al., 2015 [\[8\]](#)) and in *P. falciparum* are dominated by GPI-anchored proteins, the most abundant being PfMSP1 and 2 (Gilson et al., 2006 [\[9\]](#)). PfMSP1 has been suggested to have a role in merozoite attachment to the RBC (Li et al., 2004 [\[10\]](#)), but recent evidence instead suggests that it has a role in merozoite rupture from schizonts and cellular interaction studies using optical tweezers do not support a substantial role for this protein in RBC binding (Das et al., 2015 [\[11\]](#); Kals et al., 2024 [\[12\]](#)), although it may have another function in invasion. These recent insights into PfMSP1 function, the best studied MSP, highlight what little is known of MSPs and their roles in parasite survival.

P. falciparum merozoite surface protein 2 (PfMSP2), a protein proposed to be essential for parasite growth (Sanders et al., 2006 [\[13\]](#)), has been of long-term interest as a vaccine candidate. PfMSP2 is a ~28 kDa intrinsically disordered protein that has conserved N- and C- termini, along with a central variable region that defines two main allelic groups, 3D7-like and FC27-like (Adda et al., 2012 [\[14\]](#)). While PfMSP2 was theorised to have a mechanical role in the early steps of invasion (Anders et al., 2010 [\[15\]](#)), there is minimal supporting evidence for this, in part due to the lack of tools to study PfMSP2 function. Studies with recombinant proteins suggest that it is likely PfMSP2 interacts with lipids and forms complexes with itself on the merozoite surface (Adda et al., 2009 [\[16\]](#); Lu et al., 2019 [\[17\]](#); Zhang et al., 2012 [\[18\]](#)). PfMSP2 peptides have also been reported to bind the RBC and inhibit *P. falciparum* growth, which would support PfMSP2 mediating merozoite-RBC interactions (Ocampo et al., 2003 [\[19\]](#)). Unlike the majority of the merozoite surface coat proteins, PfMSP2 is not present in other malaria parasites that infect humans and has been postulated to have evolved specifically in the *Laverania* subgenus of *Plasmodium*, which includes *P. falciparum* (Black et al., 2002 [\[20\]](#)).

PfMSP2 has been trialled in a vaccine (Combination B) combined with fragments of PfMSP1 and PfRESA. A Phase 1/2b trial of this vaccine showed efficacy in reducing the parasite burden in children, which was specific to the PfMSP2 3D7-like allele used in the vaccine formulation (Genton et al., 2003 [\[21\]](#)). Vaccine-induced antibodies from clinical trials or experiments in animals have little or no inhibitory activity in standard growth inhibition assays (Boyle et al., 2015 [\[22\]](#); McCarthy et al., 2011 [\[23\]](#)), but did show functional activity through the recruitment of complement, promoting opsonic phagocytosis and antibody-dependent cellular inhibition (Boyle et al., 2015 [\[22\]](#); Feng et al., 2018 [\[24\]](#); McCarthy et al., 2011 [\[23\]](#)). Naturally acquired antibodies are also known to target PfMSP2 and have been linked to protection (Fowkes et al., 2010 [\[25\]](#); Osier et al., 2010 [\[26\]](#); Stanisic et al., 2009 [\[27\]](#); Zerebinski et al., 2024 [\[28\]](#)). Studies have highlighted that it is the Fc-mediated

functional activities of these naturally acquired *PfMSP2* antibodies, as opposed to direct blocking of *PfMSP2* protein function, that correlates best with protection (Boyle et al., 2015; Osier et al., 2014; Reiling et al., 2019). While *PfMSP2* is a key target of naturally acquired malaria immunity, little is known about the function of *PfMSP2*, hindering its advancement as a potential vaccine candidate.

Here we take a reverse genetics approach to assess the function of *PfMSP2*, quantifying the impacts of *PfMSP2* deletion on parasite invasion and phenotype. Given our findings that deletion of *MSP2* did not impact invasion *in vitro*, we tested the alternative hypothesis that merozoite surface proteins may function to modulate susceptibility of merozoites to inhibitory antibodies.

Results

Avian malaria MSP2-like proteins are structurally similar to *Laverania* MSP2s and show that MSP2 did not arise exclusively in this clade

Genome annotations of *pfmsp2* show it lies between the merozoite surface protein *pfmsp5* (PF3D7_0206900) and *pfmsp4* (PF3D7_0207000), and the conserved purine biosynthesis enzyme *adenylosuccinate lyase* (ADSL, PF3D7_0206700) (Figure 1A) (PlasmoDB; (Aurrecoechea et al., 2009)), with this arrangement previously thought to be restricted to the *Laverania* clade which includes parasites that infect great apes and *P. falciparum* (Black et al., 2002; Ferreira et al., 2015). As described by Escalante et al. (2022), we found the genomes of the two annotated avian *Plasmodium* spp. available on PlasmoDB, *P. relictum* and *P. gallinaceum*, to also have a putative gene between ADSL and *MSP5* with structural similarities to *Laverania* MSP2s. The reduced length of the *P. relictum* (PRELSG_0415300) and *P. gallinaceum* (PGAL8A_00017700) MSP2-like proteins compared to those of *Laverania* parasites is accounted for by the absence of any extended central domains of the two avian malaria parasites (Supplementary Figure 1).

Given the estimated that ~10 million years of evolution separate the avian malaria parasites *P. relictum* and *P. gallinaceum* from the *Laverania* (Böhme et al., 2018), we explored whether similarities in amino acid sequence and predicted conformation would support a shared lineage and functional properties of these putative MSP2s. Although *PfMSP2* is generally considered a disordered protein with limited conservation, the hydrophobic N- and C-terminal sequences that correspond to secretion signals (SP) and GPI-attachment signals, as well as short regions downstream and upstream of these signal sequences respectively, show significant conservation across *P. falciparum* MSP2s. Here we compared the sequence similarity of these conserved regions between the *Laverania* and avian MSPs and found consistently high levels of conservation for the putative SP (>89% similarity) (Supplementary Figure 1), the 24 amino acid N-terminal conserved region (78% similarity) that has a propensity for alpha-helical structure and fibril formation (Yang et al., 2010) (Supplementary Figure 2), the 50 amino acid residue C-terminal conserved region (62% similarity) (Supplementary Figure 3) and the C-terminal GPI signal sequence (>79%), including the presence of short side chain amino acids that mark predicted GPI cleavage sites (Eisenhaber et al., 1999)(Smythe et al., 1991)(Gilson et al., 2006) (Supplementary Figure 1). *Laverania* MSP2s contain two C-terminal cysteine residues that form an intramolecular disulphide bond (Zhang et al., 2008). This arrangement of 2 C-terminal cysteines is found in *P. gallinaceum* MSP2, but not *P. relictum* which has only one (Supplementary Figure 3). Overall, these N and C-terminal conserved regions show broad conservation in properties across the *Laverania* and avian malarias which could contribute to a conserved sub-cellular localisation.

Characteristic of *P. falciparum* MSP2s is the dimorphic central variable repeat region that are grouped as 3D7 or FC27-like amino acid sequence structures. A defining feature of the 3D7-like MSP2 family are 4 or 8-mer tandem repeats consisting of small amino acids (for 3D7: Gly, Gly, Ser, Ala) (MacRaild et al., 2015; Smythe et al., 1991). The FC27-like MSP2 family also has repeats of 12 or 32 amino acids in length, but these do not resemble those of the 3D7-like family. We found that the non-*P. falciparum* *Laverania* and avian malaria MSP2 sequences also contained >75%

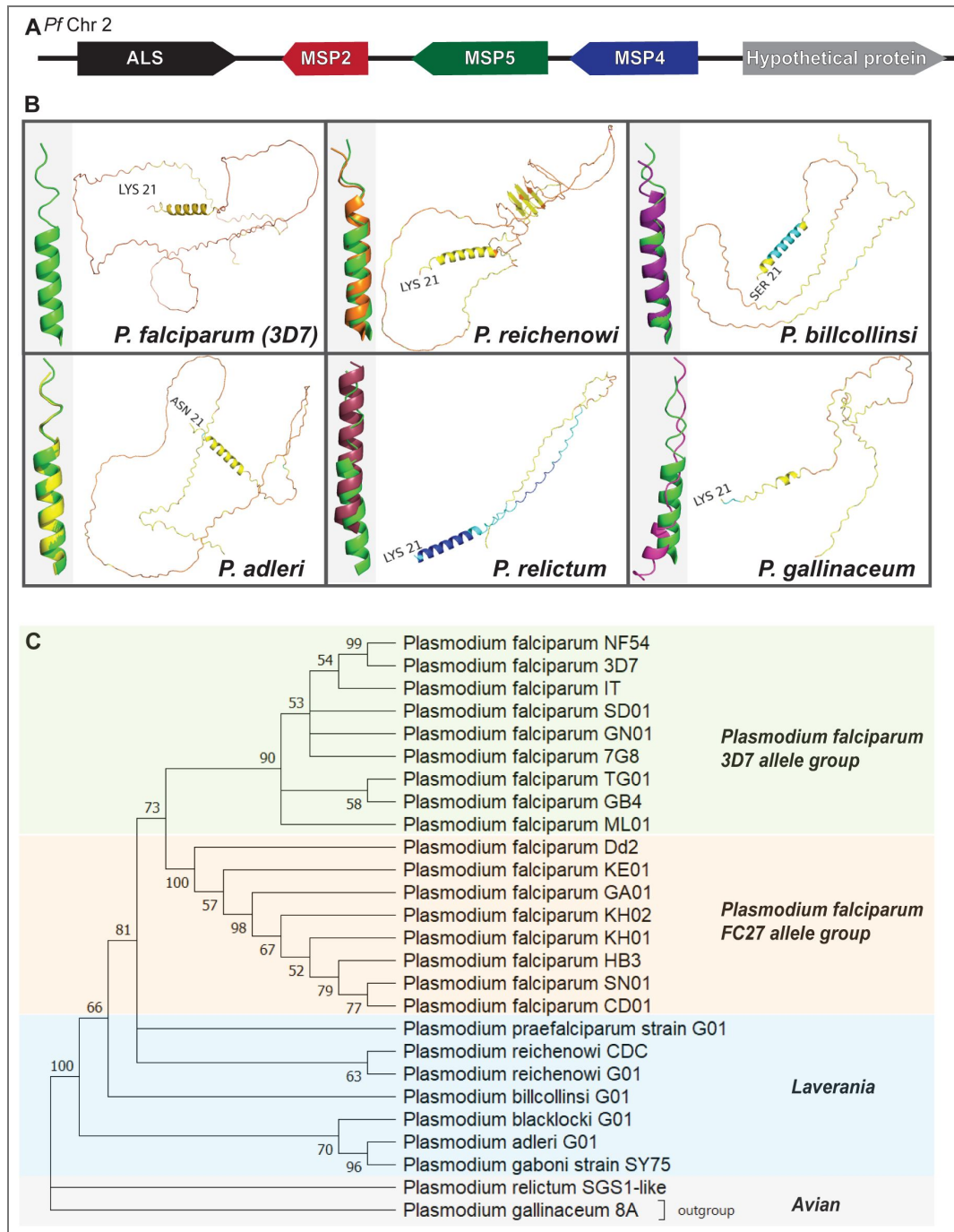


Figure 1. Presence and structure of MSP2 across different *Plasmodium* species.

(A) Schematic of the gene arrangement surrounding *msp2* on *P. falciparum* chromosome 2. **(B)** AlphaFold2 structural predictions of example *Laverania* (*P. falciparum* (3D7), *P. reichenowi*, *P. billcollinsi*, *P. adleri*) and the two avian (*P. relictum*, *P. gallinaceum*) malaria species. The N-terminal signal peptide and C-terminal GPI anchor sequences were removed before the proteins structure was predicted. The most N-terminal amino acid is indicated. Colours represent the predicted local distance difference test (pLDDT) scores, with dark blue representing very high confidence (>90%), light blue high confidence (90 to >70), yellow low confidence (70 to 50) and orange very low confidence (<50). An enlarged modelled structure of the N-terminal helical region is provided for *P. falciparum* 3D7 MSP2 on the left of the panel (grey shading), and also for the other examples of *Laverania* and avian MSPs (various colours) superimposed on the structure of *P. falciparum* 3D7 MSP2 (green). **(C)** Maximum likelihood tree showing relationship of MSP2 protein sequences found in different *P. falciparum* isolates and other *Plasmodium* species. Grouping of sequences from *P. falciparum* isolates into two main allele types can be seen as well as the separation of the *Laverania* and avian malaria species into their expected groups. Tree robustness tested by bootstrap.

small amino acids (some of: Gly, Ser, Ala, Thr) repeats, but the exact number (4 to 8 repeats), length (3 to 7 amino acids) and the level of conservation between repeats differed between and within species (Supplementary Figure 4 [\[1\]](#)). Broadly speaking, these *Laverania* and avian malaria MSP2 repeats had a 3D7-like repeat structure, with no example of a FC27-like repeat structure identified even with an additional nine *Laverania* MSP2s from the NCBI experimental database compared (data not shown). This comparison highlights the surprising finding that a region of known dimorphic variability in *P. falciparum* MSP2 is likely to retain functional constraints that favour small amino acid rich repeats immediately downstream of the N-terminal conserved region.

Extensive studies on recombinant proteins of the *P. falciparum* 3D7 and FC27 variants of PfMSP2 have shown this merozoite surface protein to be intrinsically disordered. Given the evidence for conservation of the primary structure of MSP2s from distantly related malaria parasites, we used AlphaFold2 predictions (Jumper et al., 2021 [\[2\]](#)) to examine whether there were likely to be also conformational similarities between *Laverania* MSP2s and those of *P. relictum* and *P. gallinaceum*. The MSP2s in the database of AlphaFold2-predicted structures retain their N- and C-terminal signal sequences, but as these are both absent from mature MSP2 on the merozoite surface they were deleted from the MSP2 sequences used for the AlphaFold2 predictions reported here. The predicted structures of the two avian parasite MSP2s, and other *Laverania* MSP2s, are very similar to that of Pf3D7 MSP2, with most of the polypeptide chains lacking predicted secondary structure and, from the colour coding, have a predicted local distance test (pLDDT) consistent with that of an intrinsically disordered protein over the majority of the proteins length (Figure 1B [\[3\]](#)). In contrast to the rest of the polypeptide, some α -helical structure was predicted in the conserved 20-residue N-terminal region of all the *Laverania* and avian MSP2s. NMR studies with recombinant Pf3D7 and PfFC27 MSP2 showed this conserved N-terminal sequence to have a propensity for forming an α -helix that was stabilized in the presence of lipid mimetics (MacRaild et al., 2012 [\[4\]](#)). The low pLDDT values are consistent with this region in the other *Laverania* and *P. gallinaceum* MSP2 having some propensity for α -helix formation, whereas higher pLDDT values suggest the possible presence of a more stable α -helix in this region of *P. relictum* MSP2.

Having established that MSP2 is present across different *Plasmodium* lineages sharing similar predicted amino acid compositions and structural features, the phylogenetic relationship between annotated *Laverania* and putative avian malaria parasite MSP2s was examined. A phylogenetic tree was generated based on alignments of representative sequences from each species. MSP2 sequences from species belonging to the *Laverania* subgenus all grouped together with two smaller clusters, corresponding to the known *Laverania* clade A and clade B species (Figure 1C [\[5\]](#)). As expected, *P. relictum* and *P. gallinaceum* MSP2 also group together. The phylogenetic tree generated based on MSP2 sequences mirrors the proposed evolution of *Plasmodium* spp. (Figure 1C [\[5\]](#)). As postulated by Escalante et al. (2022) [\[6\]](#), these data suggest that MSP2 was likely present in the ancestral *Plasmodium* parasite prior to the split of the mammalian infecting *Plasmodium* spp. from those that infect birds and lizards. Given the shared amino acid sequence and predicted conformational similarities, including the high levels of intrinsically disordered sequence outside of the N- and C-terminal regions, the *P. relictum* and *P. gallinaceum* MSP2-like proteins appear likely to contain enough properties to potentially be functionally equivalent to *Laverania* MSP2s. This conservation of MSP2-like proteins across different species suggests it has an important function.

Loss of PfMSP2 does not noticeably impact growth or invasion of different *P. falciparum* lines *in vitro*

Given previous unsuccessful attempts to disrupt *pfmsp2* (Sanders et al., 2006 [\[7\]](#)), and its high abundance on the merozoite surface (Gilson et al., 2006 [\[8\]](#)), PfMSP2 has been traditionally viewed as an essential *P. falciparum* protein with an essential function in merozoite invasion. We employed CRISPR-Cas9 gene editing to determine whether the reported inability to knock-out *pfmsp2* *in vitro* was a result of this protein being essential for parasite survival or because of the lower efficiency of the previously used gene-editing system (Sanders et al., 2006 [\[7\]](#)). Unexpectedly,

we confirmed successful disruption of *pfmsp2* by replacing the coding sequence between 132 bp and 819 bp of the gene with a hDHFR drug selection cassette in the 3D7 *P. falciparum* laboratory-adapted line (Figure 2A and B), resulting in *Pf3D7 ΔMSP2* parasites. We confirmed that *PfMSP2* was no longer expressed in *Pf3D7 ΔMSP2* parasites by western blot, which detected a *MSP2* band around 50 kDa in *Pf3D7* WT parasite material but not in *Pf3D7 ΔMSP2* parasites (Figure 2C). Similarly, immunofluorescence microscopy using anti-*PfMSP2* rabbit polyclonal antibodies showed the expected surface localisation of *MSP2* on *Pf3D7* WT merozoites but not on *Pf3D7 ΔMSP2* parasites (Figure 2D). We next assessed whether knock-out of *PfMSP2* impacted on parasite growth over one or two cycles of development *in vitro* and found that deletion of *Pf3D7 MSP2* did not cause any significant change in growth (Figure 2E and F).

As *PfMSP2* has been considered, to this point, as essential for *P. falciparum* growth *in vitro* (Sanders et al., 2006), we investigated whether *PfMSP2* could also be removed from *PfDd2*, an isolate of *P. falciparum* that differs from 3D7 in geographical origin, RBC receptor usage and allelic type of *pfmsp2*. Using CRISPR-Cas9 we successfully knocked-out *mSP2* in *PfDd2* (Figure 3A and B) and confirmed the absence of the protein in *PfDd2 ΔMSP2* parasites using western blot (Figure 3C). As we found for *Pf3D7*, deletion of *PfDd2 mSP2* did not result in a significant impact on parasite growth over one or two cycles of parasite replication (Figure 3D).

Given the abundance of *PfMSP2* on the merozoite surface, we postulated that loss of this protein may impact on the timing of merozoite invasion of the RBC, which is not reflected in reduced growth in *in vitro* cultures. Previous studies using live-cell microscopy have reported that merozoites first contact the RBC and form weak initial interactions. The merozoite then reorients to bring apical organelles into position to release their contents onto the RBC, which then leads to formation of the tight junction, merozoite invasion through the tight-junction, formation of the parasitophorous vacuole and resealing of the RBC membrane post-entry (Weiss et al., 2015). Given its surface localization, *PfMSP2* has been speculated to act in the early phase of merozoite attachment. To assess whether knock-out of *pfmsp2* impacted on the progression of invasion, we used live-cell microscopy of *PfDd2 ΔMSP2* parasites and compared the timing and key steps of invasion to *PfDd2* WT parasites *in vitro*. This analysis showed that, although there was a trend for *PfDd2 ΔMSP2* knock-out parasites to have a higher mean time to attach to the RBC, as well as for the length and strength of RBC deformation, these trends did not reach significance. For those merozoites that did invade the RBC, on average it took less time for *PfDd2 ΔMSP2* knock-out parasites to invade than *PfDd2* WT, but this again did not reach significance (Figure 3E-H). Together these data show *PfMSP2* is not essential for blood-stage replication *in vitro* in two *P. falciparum* laboratory isolates from different geographical regions and knock-out of *PfMSP2* does not seem to significantly impact parasite growth or merozoite invasion *in vitro*. However, its conservation across diverse *Plasmodium* spp. suggests it does play an important function or is advantageous for parasite survival.

Impact of *MSP2* gene deletion on gene expression and invasion pathway usage

Because of the close proximity of the *pfmsp2*, 4 and 5 genes on chromosome 2, similarities in their structure (e.g. GPI-anchor intrinsically disordered protein) and shared merozoite surface localization, we used qPCR to assess whether knock-out of *Pf3D7 MSP2* impacted the expression levels of *Pf3D7 MSP4* or *MSP5*. We found no change in *MSP4* and *MSP5* expression levels between *Pf3D7 ΔMSP2* parasites and *Pf3D7* WT parasites (Figure 4A). As expected, *PfMSP2* was not detected in the *Pf3D7 ΔMSP2* parasites.

Secreted *P. falciparum* antigens belonging to the erythrocyte binding antigen (EBA) and reticulocyte binding homologue (Rh) families are known to facilitate host-cell invasion through specific receptors on the RBC surface. Reliance on specific EBA or Rh proteins, or their cognate host-cell receptor, define different invasion pathways that can be crudely described by the sensitivity of merozoite invasion to cleavage of RBC surface proteins with trypsin, chymotrypsin and neuraminidase (Baum et al., 2005; Duraisingh et al., 2003; Orlandi et al., 1992). When

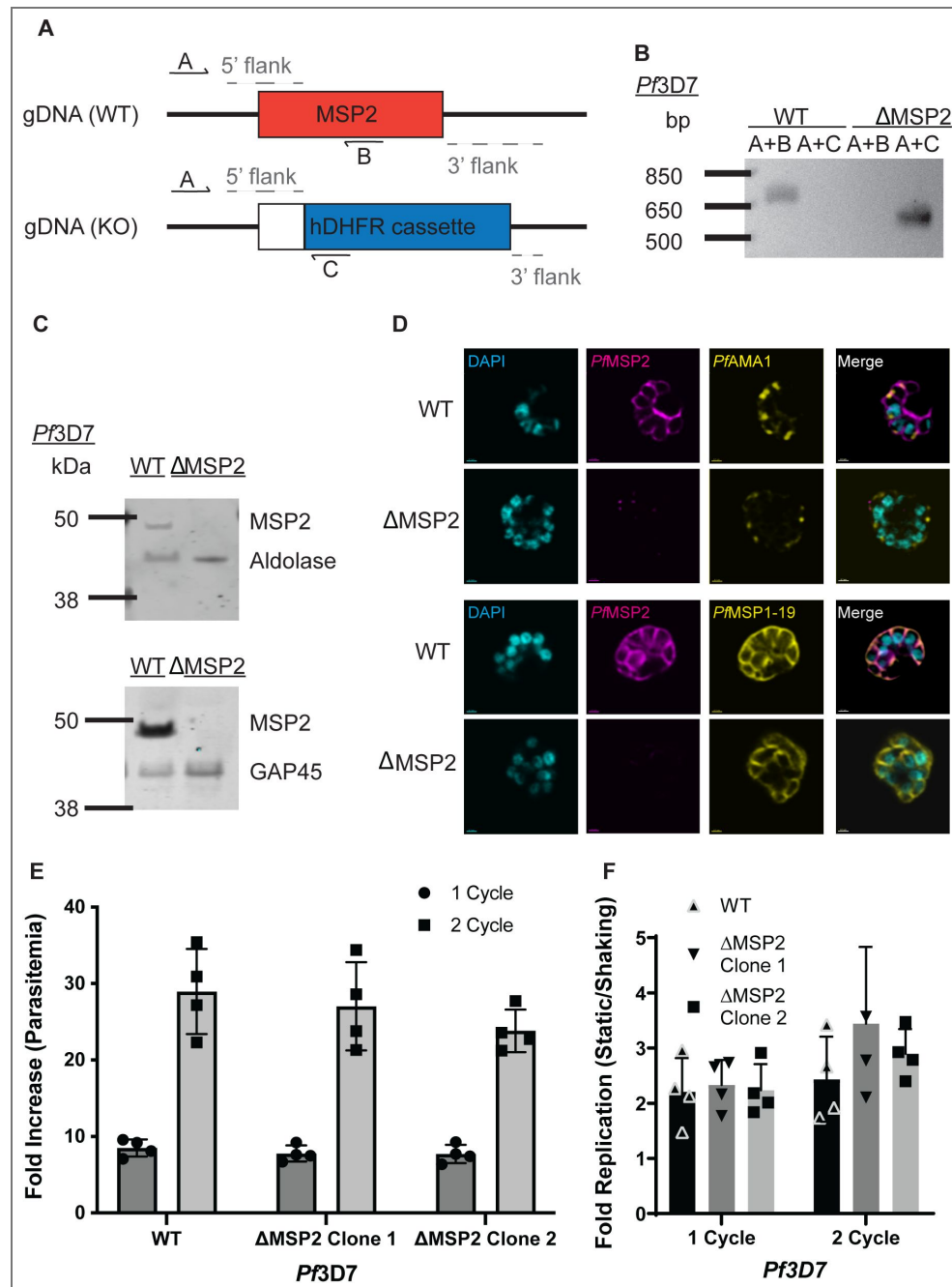


Figure 2. *PfMSP2* is not essential for growth *in vitro* of *Pf3D7* blood stage parasites.

(A-B) Schematic and agarose gel showing integration of knockout construct (band A+C) in the *msp2* gene locus and absence of the *msp2* sequence from the *Pf3D7* ΔMSP2 line (band A+B). (C) Western Blot of late schizont protein extracts confirms no *PfMSP2* is expressed in the *Pf3D7* ΔMSP2 line. *PfMSP2* detected by anti-*PfMSP2* 2F2 3D7 mAb with *PfAldolase* (upper blot) or *PfGAP45* (lower blot) serving as loading and stage of expression controls respectively. Representative image shown. (D) Distribution of key merozoite surface proteins in the presence or absence of *PfMSP2* was visualised by immunofluorescence. *PfMSP2* (magenta), the nucleus stained by DAPI (cyan) and *PfAMA1* (yellow, top two rows) or *PfMSP1-19* (yellow, bottom two rows), and the coloured merge of the preceding panels. Scale bar = 0.7 μm. Representative images shown from a minimum of 10 schizonts imaged per condition. (E-F) Growth of *Pf3D7* WT compared to *Pf3D7* ΔMSP2 *P. falciparum* parasites, measured as fold increase in parasitaemia, over one (48 hrs) or two (96 hrs) cycles in either standard (still- (E)) or shaking (F) conditions, with no measurable difference between parasite growth rates seen between standard or shaking conditions. Parasitaemia was determined by flow cytometry at the start and end to calculate fold increase. Graph displays mean ± S.D. of three independent experiments performed with technical triplicates. Significance determined by unpaired t-test with $p < 0.05$ deemed significant.

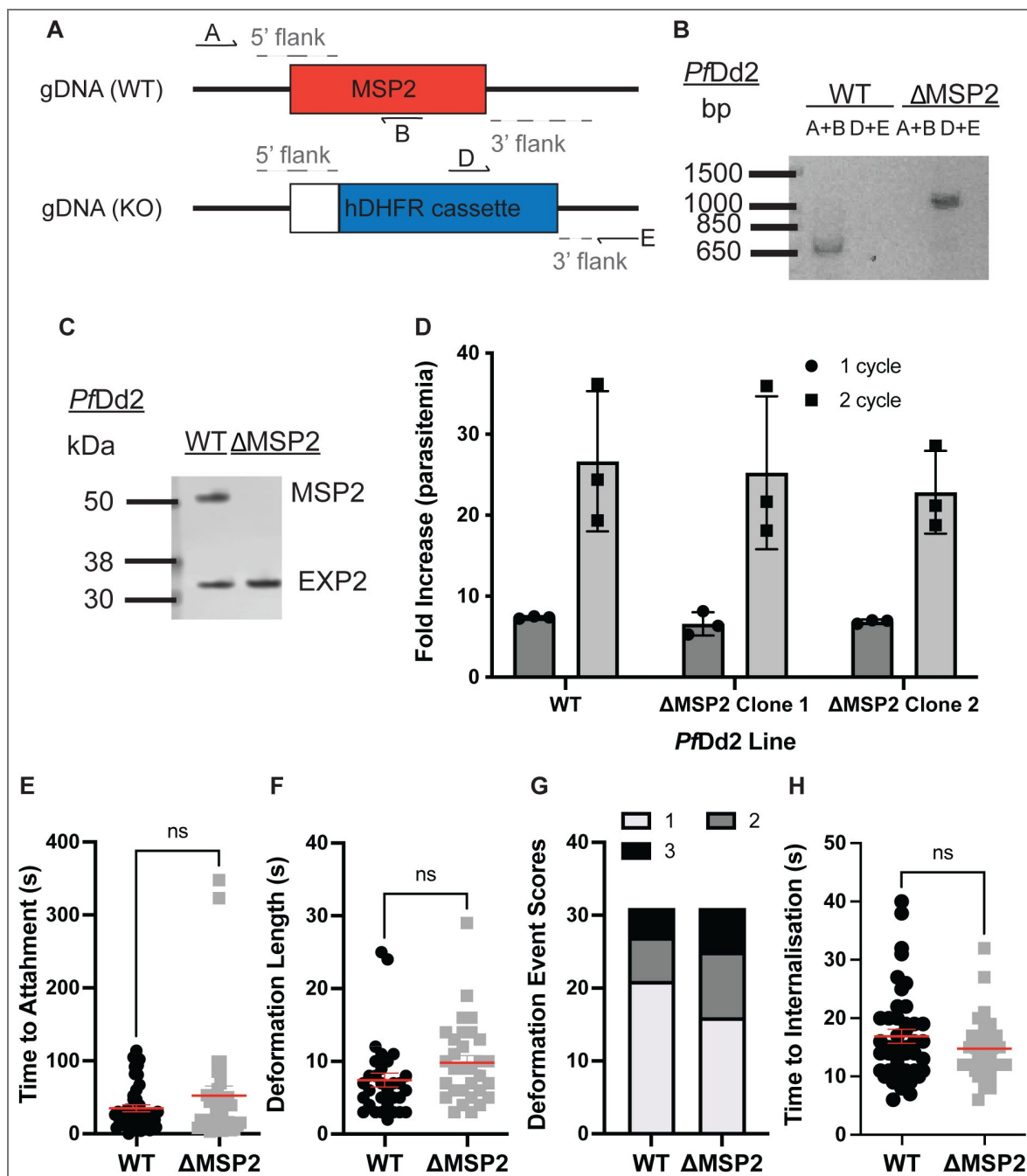


Figure 3. *PfDd2* does not require *MSP2* for asexual growth *in vitro*.

(A–B) Successful integration of KO construct (schematic in A) into the *msp2* gene locus of *PfDd2* Δ *MSP2* was confirmed by PCR of genomic DNA (primers A+B amplify WT locus, primers D+E amplify integrated KO construct). (C) Loss of *PfMSP2* expression was demonstrated by western blot of schizont protein extract with *PfMSP2* detected by anti-*PfMSP2* FC27 and anti-*PfEXP2* as loading control. Representative image shown. (D) Growth of *PfDd2* WT *P. falciparum* parasites and *PfDd2* Δ *MSP2* over one (48 hrs) or two (96 hrs) cycles. Parasitaemia was determined by flow cytometry at the start and end to calculate fold increase. Graph displays mean $\pm$ S.D. of three independent experiments performed with technical triplicates. (E–H) Key parameters of merozoite invasion were measured for both *PfDd2* WT and *PfDd2* Δ *MSP2* parasites that had successfully invaded a RBC using live cell imaging of merozoite invasion. Time to merozoite attachment to RBCs (E), length (F) and strength (G) of RBC deformation, and time to complete merozoite invasion (H) were measured by live microscopy. Deformation scores are as defined by Weiss et al (2015) [1], with 1 = weak deformation of the RBC membrane at the point of contact, 2 = strong deformation leading to the RBC membrane extending up the sides of the merozoite and changes in RBC membrane curvature beyond the point of contact and 3 = extreme deformation indicated by the merozoite being deeply embedded in the RBC membrane and strong deformation of the RBC well beyond the point of contact. A minimum of 30 invading merozoites for each line were captured and assessed. Significance determined by unpaired t-test with $p < 0.05$ deemed significant.

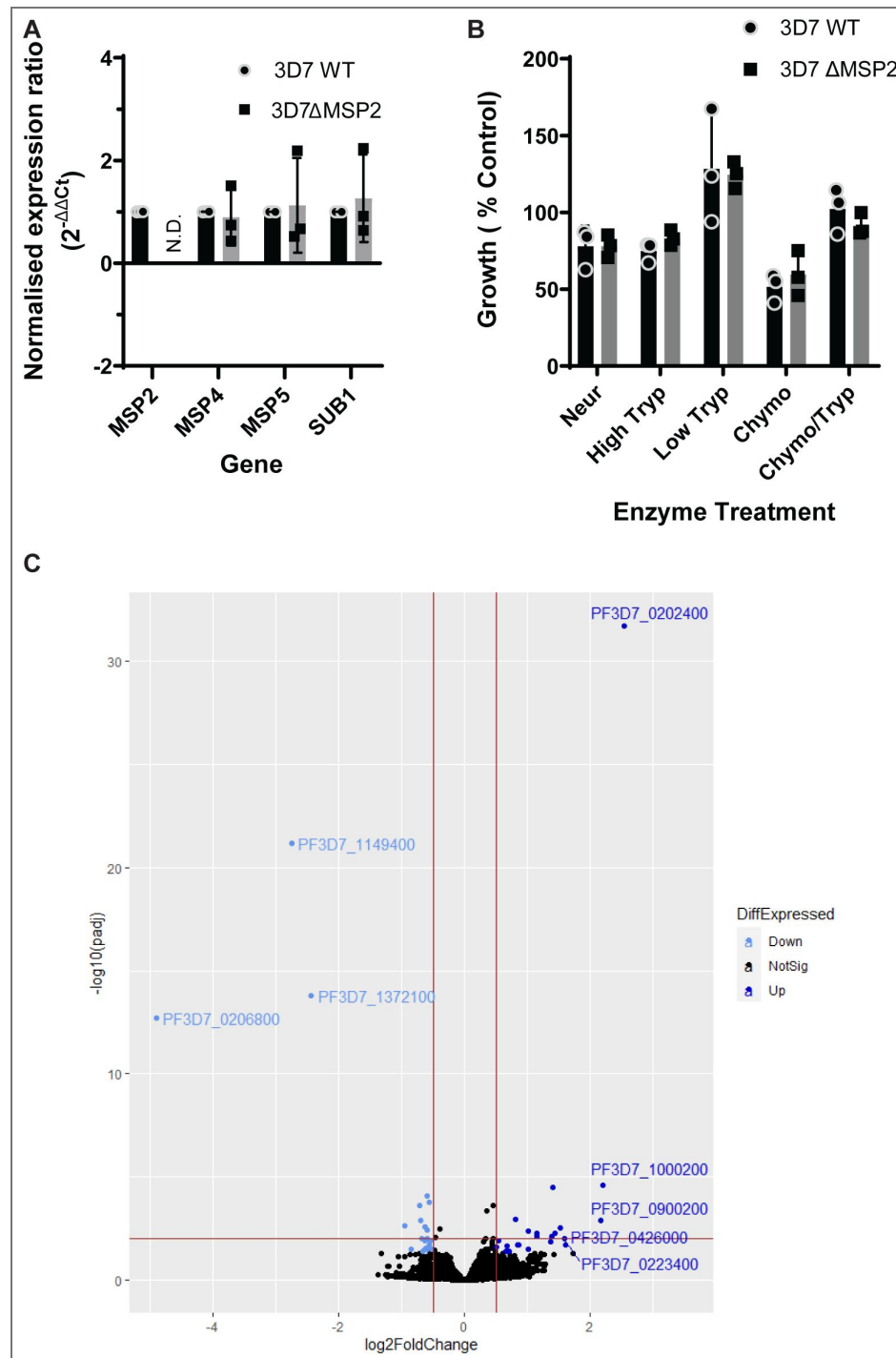


Figure 4. The impact of the loss of *PfMSP2* on expression of known merozoite invasion genes and invasion pathway utilization.

(A) Impact of *PfMSP2* KO on schizont transcript abundance was assessed by qPCR for genes located in proximity to *Pfmsp2* on chromosome 2. Changes in expression between *Pf3D7* WT and *Pf3D7* Δ MSP2 parasites was determined by qPCR relative to *pfaldolase* expression with *pfsub1* serving as a schizont stage control. Graph displays mean $\pm$ S.D. of three independent RNA harvests. **(B)** Selective enzymatic cleavage of key RBC receptors showed no difference in invasion preference between *Pf3D7*MSP2 WT and *Pf3D7* Δ MSP2 parasites. Parasitaemia was determined by flow cytometry and compared to growth in non-treated control RBCs. Graph displays mean $\pm$ S.D. of three independent experiments. **(C)** Log₂(fold change) for differentially expressed genes, including multigene families, between the transcriptome of *Pf3D7* WT and *Pf3D7* Δ MSP2 schizonts. Plot represents the results for one of four independent schizont RNA harvests for *Pf3D7* WT and *Pf3D7* Δ MSP2 parasites and red lines differentiate genes with a log₂ (fold change) > 0.5 and < -0.5 with adjusted p-value < 0.01. Genes shaded blue represent those genes that were found to have an average log₂ (fold change) > 0.5 (dark blue) or < -0.5 (light blue) across the four replicate samples compared. Significance determined as below $p < 0.05$ after correction for multiple testing.

we assessed whether *PfMSP2* knock-out impacted merozoite invasion pathway usage, we found that *Pf3D7* Δ MSP2 parasites grew equally as well as *Pf3D7* WT parasites in all enzyme-treated RBCs (Figure 4B [↗](#)).

To investigate whether *PfMSP2* knock-out caused any changes in gene-expression of proteins that may have a role in merozoite invasion and possibly compensate for loss of *PfMSP2*, we undertook RNA sequencing and differential gene expression analysis of *Pf3D7* Δ MSP2 and *Pf3D7* WT parasites. As expected, the *PfMSP2* transcript, which was targeted for removal by our CRISPR-Cas9 strategy, was not detected in the *Pf3D7* Δ MSP2 parasites between 132 bp and 819 bp (Supplementary Figure 5 [↗](#)). After removal of genes belonging to variant surface antigen families from the data, eight transcripts were found to have a log2fold expression increase above 1; none of these proteins are annotated as a merozoite surface or secreted protein linked to merozoite invasion (Figure 4C [↗](#), Table 1 [↗](#)). Five of these eight upregulated proteins are of unknown function and expressed at schizont stages, and only one (*Pf3D7_0909100*) has a transmembrane domain that could possibly anchor it to a membrane surface as the GPI anchor does for *PfMSP2* (Table 1 [↗](#)). Three proteins were found to have a log2fold expression of <-1, indicating a significant reduction in expression with *PfMSP2* knock-out, with *PfMSP2* being one of these and the other two predicted to be exported proteins (Figure 4C [↗](#), Table 1 [↗](#)). While these data do not exclude any of the *Plasmodium* proteins with unknown function that are upregulated in *Pf3D7* Δ MSP2 parasites as having a role that compensates for the loss of *PfMSP2*, only one is predicted to have a transmembrane domain that would bind it to a membrane and no protein identified as up or down-regulated with *PfMSP2* knock-out has previously been linked to merozoite invasion. Therefore, *PfMSP2* knock-out does appear to have led to changes in gene expression, but does not appear to have a significant impact on the regulation of known merozoite surface or secreted protein gene-expression that would suggest loss of *PfMSP2* is compensated for by another protein.

Impact of *PfMSP2* disruption on antibodies and inhibitors targeting secreted surface exposed antigens

Merozoites are key targets of antibodies during malaria infection with antibodies either able to directly block protein function and thus merozoite invasion, or recruit effectors such as complement leading to the destruction of the merozoite (Beeson et al., 2016 [↗](#)). Given the prominence of *PfMSP2* on the merozoite surface we asked the question; how does the loss of *PfMSP2* affect antibody efficacy against merozoites?

PfEBAs and *PfRh*s moderate early steps in invasion with a degree of redundancy between these proteins. Several EBAs and Rhs can be targeted with invasion inhibitory antibodies and are targets of acquired human antibodies (Persson et al., 2013 [↗](#), 2008 [↗](#); Richards et al., 2013 [↗](#)). We found no significant difference in the ability of purified rabbit immunoglobulin raised against single (*PfRh2b*) or combinations of *PfEBAs* and *PfRh*s (*PfEBA175/PfRh2b*, *PfEBA175/PfRh4*, *PfEBA175/PfRh2b/PfRh4*) to inhibit growth of *Pf3D7* WT or *Pf3D7* Δ MSP2 parasites (Figure 5A [↗](#)). Similarly, knock-out of MSP2 in *PfDd2* did not significantly change the parasites sensitivity to rabbit antibodies raised against isogenic W2mef EBA175 (Figure 5B [↗](#)). Broadly speaking, these results mirror what was seen with selective cleavage of the RBC ligands of *PfEBAs* and *PfRh*s using enzyme treatment of RBCs (Figure 4B [↗](#)), with no evidence that loss of *PfMSP2* changed either the importance of these secreted merozoite antigens during invasion or their sensitivity to antibodies targeting different antigens.

The interaction of *PfRh5* with RBC surface receptor basigin is a critical, non-redundant interaction for *P. falciparum* merozoite invasion and a known target of direct invasion-inhibitory antibodies (Alanine et al., 2019 [↗](#); Crosnier et al., 2011 [↗](#); Douglas et al., 2011 [↗](#)). *PfRh5* was the initial member of the PCRCR complex to be described, with subsequent studies demonstrating that antibodies against other members of this complex could also block merozoite invasion of the RBC (Sally et al., 2022 [↗](#)). Given the importance of these secreted antigens to current vaccine development efforts, we next investigated whether loss of *PfMSP2* impacted on the parasite's sensitivity to antibodies targeting these antigens. There was no difference in sensitivity between *Pf3D7* WT and *Pf3D7* Δ MSP2 parasites when treated with invasion-inhibitory rabbit antibodies

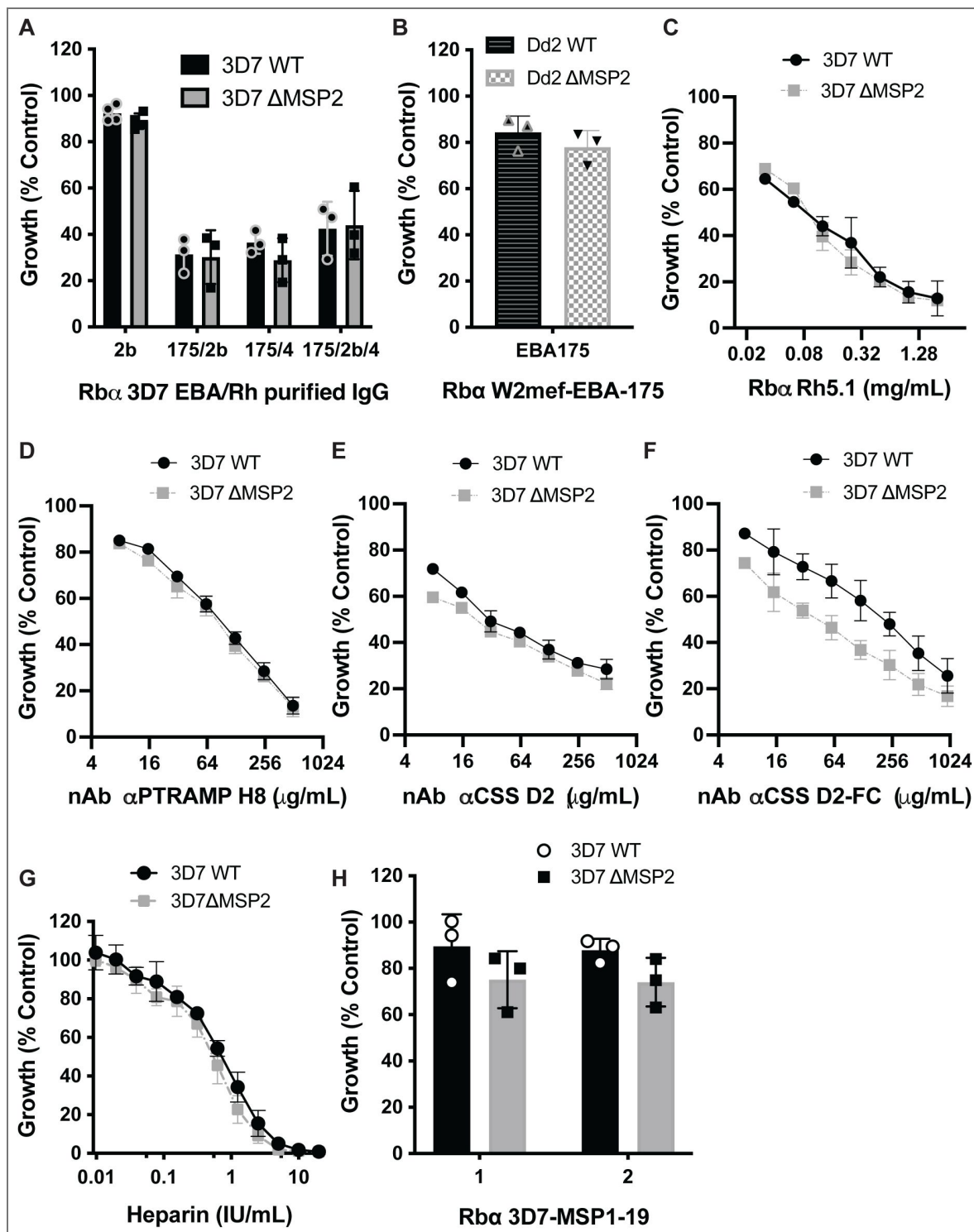


Figure 5. Impact of *PfMSP2* removal on efficacy of antibodies targeting other merozoite surface-exposed antigens.

Changes in antibody efficacy in the absence of *PfMSP2* was assessed by measuring changes in antibody invasion inhibition and subsequent growth compared to growth in the absence of antibody for both *P. falciparum* WT and ΔMSP2 parasites over 2 cycles. **(A)** Rabbit (Rb) IgG raised against merozoite antigens of the *Pf3D7* EBA/Rh family. **(B)** Rabbit sera raised against *PfDd2* EBA175. **(C)** Rabbit IgG raised against *Pf3D7* Rh5. **(D)** Nanobody (nAb) to *Pf3D7* PTRAMP. **(E-F)** Nanobody and Fc-tagged nanobody to *Pf3D7* CSS. **(G)** The invasion inhibitory glycosaminoglycan heparin. **(H)** Rabbit sera raised against *Pf3D7* MSP1-19 (different vaccinated rabbit sera identified by numbers). Graph displays mean ± S.D. of three different experiments. Significance was determined by unpaired t-test when only a single concentration point was tested and for IC₅₀ comparisons an extra Sum-of-Squares F Test (best-fit LogIC₅₀) was performed with $p < 0.05$ deemed significant.

targeting *PfRh5* (Figure 5C) or nanobodies targeting PCRCR complex component *PfPTRAMP* (H8) (Figure 5D). Nanobodies targeting the PCRCR complex component CSS (D2) showed a slight, but non-significant increase in invasion-inhibitory activity in the absence of *PfMSP2* (1.4-fold increase; IC₅₀ value for *Pf3D7* WT is 57 µg/mL; IC₅₀ value for *Pf3D7* Δ*MSP2* is 42 µg/mL; *p*=0.3; Figure 5E). However, the *Pf3D7* Δ*MSP2* specific increase in invasion-inhibitory activity was amplified with the addition of the human Fc domain to the anti-CSS nanobody (D2-Fc) to be 3-fold greater for *Pf3D7* Δ*MSP2* parasites (IC₅₀ *Pf3D7* WT 138 µg/mL; IC₅₀ *Pf3D7* Δ*MSP2* 46 µg/mL; *p*=0.0001; Figure 5F).

The glycosaminoglycan heparin is a specific inhibitor of invasion and has been observed to bind to the 42 kDa C-terminal region of *PfMSP1* the most abundant GPI-anchored protein on the merozoite surface, marking this as a possible mechanism for its invasion inhibitory activity (Boyle et al., 2010a). However, the highly charged state of heparin means that it could block invasion through binding additional proteins. When tested against *Pf3D7* WT and *Pf3D7* Δ*MSP2* parasites, we observed a small increase in inhibitory potency of heparin for parasites lacking *PfMSP2* (1.4-fold; IC₅₀ *Pf3D7* WT 0.65 IU/mL; IC₅₀ *Pf3D7* Δ*MSP2* 0.47 IU/mL; *p*=0.0003; Figure 5G). We next tested rabbit antibodies raised against the C-terminal 19 kDa fragment of *PfMSP1*. We found a small but non-significant trend towards increased inhibition of growth in parasites that lacked *PfMSP2* across repeat experiments (Figure 5H). These data suggest that loss of *PfMSP2* can impact on the invasion-inhibitory potency of some antibodies that target secreted or surface antigens.

Loss of *PfMSP2* consistently potentiates AMA1 invasion inhibitory antibodies

Subsequent to *PfRh5*-basigin binding in the steps of merozoite invasion is the formation of the tight-junction between *PfAMA1*, secreted from the micronemes onto the merozoite surface, and the rhoptry neck protein *PfRON2*, which is inserted into the RBC membrane (Lamarque et al., 2011; Srinivasan et al., 2011). In lines where *PfMSP2* was disrupted, we found a consistent potentiation of antibodies that had anti-*PfAMA1* invasion-inhibitory activity. Invasion-inhibitory polyclonal rabbit serum antibodies and purified immunoglobulin raised against *PfAMA1* of *Pf3D7* and *PfW2mef* (isogenic to *PfDd2*), which express different alleles of both *PfMSP2* and *PfAMA1*, were found to be more inhibitory with *PfMSP2* knock-out (Figure 6A and B). The polyclonal antibody with the greatest potentiation of inhibitory activity was an IgG preparation purified from a rabbit antiserum raised against *PfW2mef* AMA1. This had a 3-fold increased potency against *PfDd2* Δ*MSP2* compared to *PfDd2* WT (IC₅₀ *PfDd2* WT 0.015 mg/mL; IC₅₀ *PfDd2* Δ*MSP2* 0.005 mg/mL; *p*<0.0001) over two cycles of growth (Figure 6B, C).

The mouse 1F9 (binds to the RON2 binding pocket and polymorphic loop 1d (Coley et al., 2007)) and rat 4G2 (binds to a conserved epitope in the ectodomain (Kocken et al., 1998)) monoclonal antibodies have been shown to target different regions of *PfAMA1* and exhibit selective inhibition against *Pf3D7* AMA1. Here we found potentiation of invasion inhibitory activity with *Pf3D7* *MSP2* knock-out for both 1F9 (2.1-fold; IC₅₀ *Pf3D7* WT 0.27 mg/mL; IC₅₀ *Pf3D7* Δ*MSP2* 0.13 mg/mL; *p*<0.0001; Figure 6D) and 4G2 (3.2-fold; IC₅₀ *Pf3D7* WT 1.15 mg/mL; IC₅₀ *Pf3D7* Δ*MSP2* 0.36 mg/mL; *p*<0.0001; Figure 6E).

We also tested the invasion inhibitory effect of i-bodies, which are smaller single-domain antibody-like molecules inspired from the shark variable new antigen receptor (V_{NAR}) (Griffiths et al., 2016, 2018). When we tested the i-body WD34 (Angage et al., 2024) which binds a highly conserved epitope that includes the *PfRON2*-binding pocket on *PfAMA1*, we observed a small potentiation of *PfAMA1* specific activity with knock-out of *PfMSP2* in *Pf3D7* (1.3-fold; IC₅₀ *Pf3D7* WT 0.012 mg/mL; IC₅₀ *Pf3D7* Δ*MSP2* 0.009 mg/mL; *p*=0.08 Figure 6F).

Given WD34 was also found to be inhibitory to growth of *W2mef* parasites (Angage et al., 2024) we also tested its activity against our *PfDd2* Δ*MSP2* parasites and found potentiation of invasion inhibitory activity with loss of *MSP2* in *PfDd2* parasites (2-fold; IC₅₀ *PfDd2* WT 0.016 mg/mL; IC₅₀ *PfDd2* Δ*MSP2* 0.008 mg/mL; *p*=0.004 Figure 6G). As was observed for the anti-CSS nanobody, addition of a human Fc domain to WD34 amplified the inhibitory phenotype, with WD34 Fc having

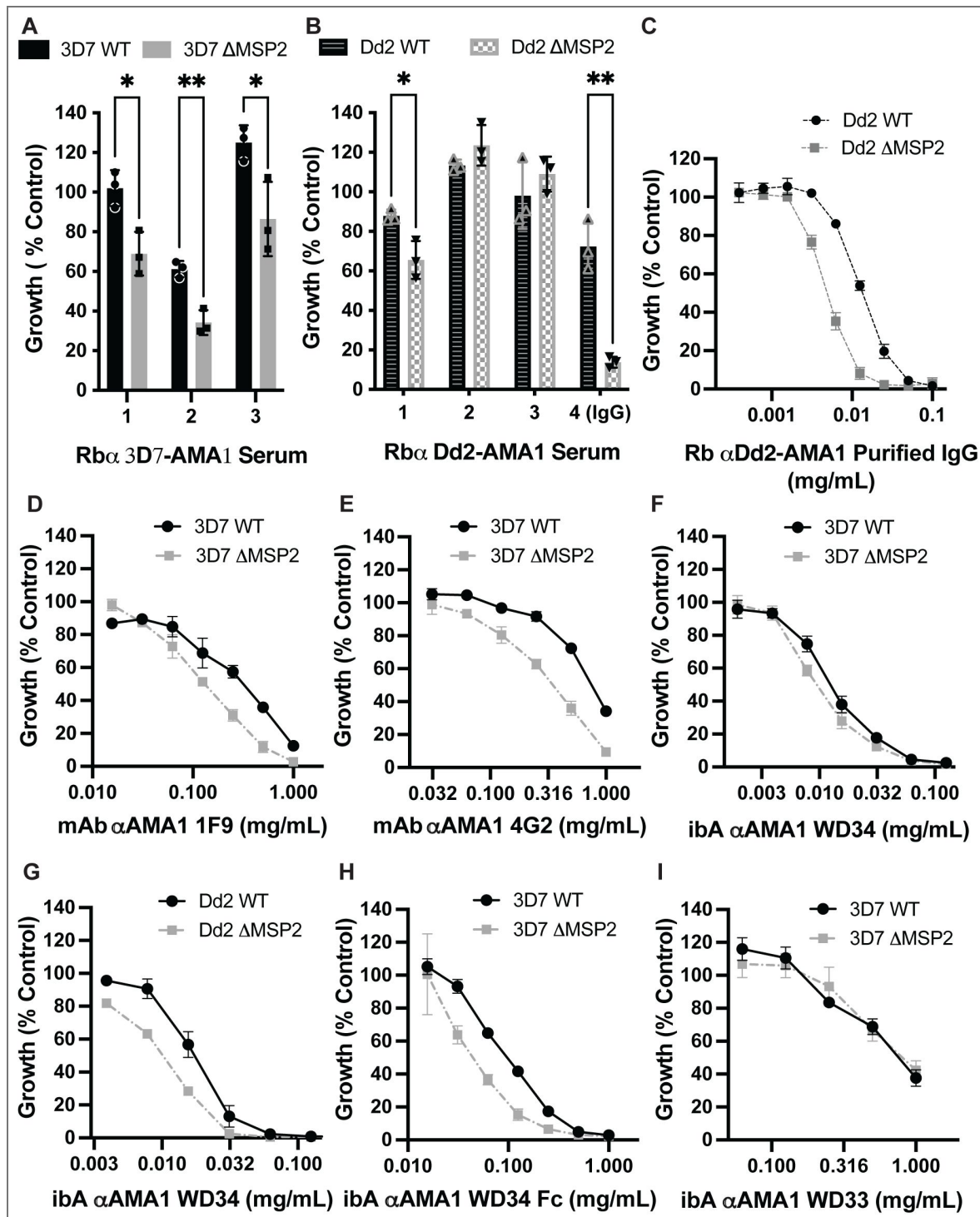


Figure 6. Absence of *PfMSP2* from the merozoite surface impacts invasion inhibition by *PfAMA1* antibodies.

Pf3D7 (A, D-F, H, I) and *PfDd2* (B, C, G) express different *PfMSP2* alleles and different *PfAMA1* alleles, yet both showed altered anti-AMA1 antibody growth inhibition in the absence of *PfMSP2*. The effect was seen with serum (A-B; different vaccinated rabbit (Rb) sera identified by numbers), purified rabbit and mouse monoclonal (mAb) antibodies (C-E) and i-bodies (iB) (F-I). Final parasitaemia was determined by flow cytometry and compared to control. Graph displays mean $\pm$ S.D. of three or four independent experiments. Significance was determined by unpaired t-test when only a single concentration point was tested and for IC_{50} comparisons an extra Sum-of-Squares F Test (best-fit $LogIC_{50}$) was performed with $p < 0.05$ deemed significant.

a 2.3-fold greater inhibitory activity against *Pf*3D7 Δ MSP2 than against *Pf*3D7 WT (IC_{50} *Pf*3D7 WT 0.1 mg/mL; IC_{50} *Pf*3D7 Δ MSP2 0.04 mg/mL; $p=0.0004$; Figure 6H). A second i-body, WD33 (Angage et al., 2024), which binds AMA1 between domain II and domain III, had very limited invasion inhibitory activity against *Pf*3D7 parasites and did not show improved potency with knock-out of *Pf*3D7 MSP2 (0.9-fold; IC_{50} *Pf*3D7 WT 1.02 mg/mL; IC_{50} *Pf*3D7 Δ MSP2 1.1 mg/mL; $p=0.8$; Figure 6I). Although the limited inhibitory activity for WD33 tagged with the Fc receptor prevented us determining an IC_{50} at the concentrations feasible to test, there was no increased inhibition of *Pf*3D7 Δ MSP2 parasites compared to the low level seen with *Pf*3D7 WT parasites (data not shown). These data suggest that the potentiation of *Pf*AMA1 targeted antibody inhibition with MSP2 knock-out is *Pf*AMA1 epitope dependent.

To account for the differences in i-body sizes, we compared the WD34 and WD34 Fc i-bodies on a molar basis and found that they had similar activities against *Pf*3D7 MSP2 WT parasites (IC_{50} 0.94 μ M and 1 μ M respectively) indicating that the Fc-tag itself did not modify epitope binding properties. Rather, it is the absence of *Pf*MSP2 that results in increased inhibition of AMA1 function by WD34, and this is further potentiated by antibody size.

In light of these results, we hypothesised that removal of *Pf*MSP2 may improve antibody access to other antigens that are targets of inhibitory antibodies. To assess whether this was the case, we first tested whether increased *Pf*AMA1 polyclonal antibody binding to the surface of merozoites could be observed for *Pf*3D7 Δ MSP2 compared to *Pf*3D7 MSP2 WT parasites using a fluorescently tagged WD34 i-body (WD34-mCherry) for quantitative immunofluorescence assays of late stage schizonts labelled with a single incubation and wash step. We observed WD34-mCherry to have a significantly higher mean fluorescence intensity for *Pf*3D7 Δ MSP2 compared to *Pf*3D7 WT (Figure 7A), supporting that the *Pf*AMA1 i-bodies may have better access in the absence of *Pf*MSP2. As a control, we tested labelling using a fluorescently tagged WD33 i-body which binds to AMA1 (Angage et al., 2024), but we found did not have improved inhibitory activity in the absence of MSP2 (Figure 6I). Mirroring the results of the growth assay, there was no difference observed for WD33-eGFP binding fluorescence for *Pf*3D7 Δ MSP2 compared to *Pf*3D7 WT parasites (Figure 7B).

We next explored whether the addition of recombinant *Pf*MSP2 could impact on the inhibitory antibody binding on-rate against substrate-bound *Pf*AMA1 recombinant protein using Surface Plasmon Resonance (SPR). For the anti-*Pf*AMA1 targeting mAb 4G2, there was a non-significant trend toward a lower on-rate in the presence of recombinant *Pf*MSP2 (Figure 7C). For the i-body WD34-Fc, there was minimal difference in detectable binding to *Pf*AMA1 in the presence or absence of *Pf*MSP2 (Figure 7C). Using a similar principle of testing antibody binding to substrate-bound *Pf*AMA1 protein in the presence, or absence of increasing concentrations of *Pf*MSP2, we assessed changes in levels of anti-*Pf*AMA1 antibody binding using an enzyme linked immunosorbent assay (ELISA). With increasing concentrations of *Pf*MSP2 added to wells coated with AMA1, we found decreased binding of anti-*Pf*AMA1 mAb 4G2 and i-body WD34-Fc (mouse Fc) by ELISA (Figure 7D). Concentrations where the *Pf*MSP2 protein was barely detectable above background using an MSP2 mAb were found to be enough to impact on the binding of *Pf*AMA1 antibodies. In contrast, increasing concentrations of the intrinsically disordered MSP4 from *P. falciparum* 3D7 (40 kDa) and the highly structured immunoglobulin domain of neural cell adhesion molecule (NCAM, CD56, 16 kDa) recombinant proteins did not impact on binding of anti-*Pf*AMA1 antibodies to recombinant AMA1 (Supplementary Figure 6). By looking at the impacts of removing *Pf*MSP2 from the merozoite surface, or adding this protein in to recombinant assays, these data provide additional support that inhibitory antibodies targeting *Pf*AMA1 bind at a higher rate in the absence of *Pf*MSP2.

Discussion

Despite decades of research and interest, the function of most merozoite surface proteins remains unclear and targeting merozoite antigens in vaccine development has not achieved high efficacy in clinical trials. Additionally, it has generally proven difficult to generate highly inhibitory antibodies through vaccination with merozoite antigens. *Pf*MSP2 is one of the most abundant

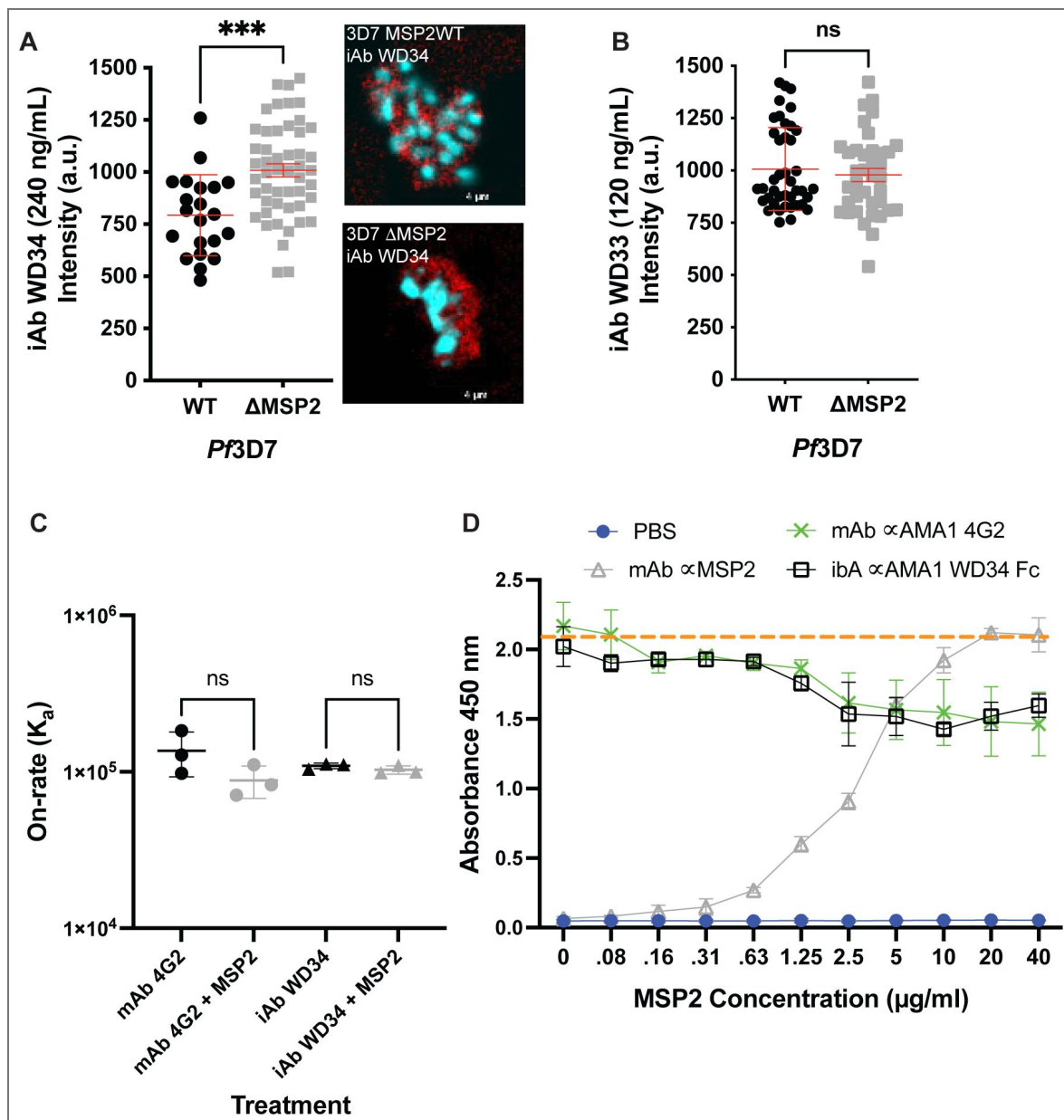


Figure 7. Quantitative fluorescence microscopy to assess whether differential binding may explain the increased potency of anti-PfAMA1 invasion-inhibitory antibodies in the absence of PfMSP2.

Fluorescence intensity of fluorescently tagged anti-PfAMA1 i-body s(iAb) WD34-mCherry (A) and WD33-eGFP (B) for both Pf3D7 WT and Pf3D7 ΔMSP2, with a representative image for i-body WD34-mCherry. Nucleus in blue, i-body signal in red. Scale bar = 4 μm. Two independent experiments were performed with significance determined by unpaired t-test with $p < 0.05$ deemed significant. The lower overall mCherry signal required a higher antibody concentration (240 ng/mL) to have a comparable intensity measure to the eGFP tagged antibody (120 ng/mL) for Pf3D7 WT merozoites. (C) Read-out of the Surface plasmon resonance (SPR) antibody on-rate (association constant) for anti-PfAMA1 mAb 4G2 and i-body WD34-Fc (mouse Fc) binding to PfAMA1 in the presence or absence of PfMSP2 protein. Data represents the mean of 3 experiments with significance determined by unpaired t-test with $p < 0.05$ deemed significant. (D) ELISA based assessment of the anti-PfAMA1 mAb 4G2 and i-body WD34-Fc antibody binding levels to recombinant Pf3D7 AMA1 in the presence or absence of recombinant Pf3D7 MSP2. PBS control demonstrates background fluorescence. Dashed orange line provides a guide for peak absorbance levels. Anti-PfMSP2 mAb shows increasing concentrations of PfMSP2 protein results in decreased binding of mAb 4G2 and i-body WD34-Fc. Data represents the mean of 3 experiments and error bars are $\pm$ S.D.

proteins on the merozoite surface, has shown promise as a vaccine candidate and is an established target of protective cytophilic antibodies (Boyle et al., 2015 [↗](#); Genton et al., 2002 [↗](#); McCarthy et al., 2011 [↗](#); Osier et al., 2014 [↗](#), 2010 [↗](#)). However, its function remains unclear. Our findings here suggest that *PfMSP2* does not have an essential role in invasion *in vitro*, but can modulate the sensitivity of merozoites to inhibitory antibodies targeting other antigens.

Despite its abundance on the merozoite surface and previous work suggesting a role in RBC invasion, we found merozoites invade normally in the absence of this protein after gene deletion. This does not exclude a role for *PfMSP2* *in vivo* where there are additional pressures, such as immune-effector mechanisms and flow dynamics, on merozoite invasion. However, given we have knocked-out *PfMSP2* from two different *P. falciparum* isolates, our findings do not currently support a major role for *PfMSP2* in the mechanics of merozoite invasion. Thus, it appears that the function of the two most abundant proteins on the merozoite surface, *PfMSP1* (Das et al., 2015 [↗](#); Kals et al., 2024 [↗](#)) and *PfMSP2*, are not directly linked to merozoite binding to the RBC and subsequent invasion. Deletion of *MSP2* did lead to significant changes in the expression of several genes, but there were no proteins with a clear link to merozoite invasion and the biological relevance of these changes is not currently understood and requires future investigation.

Both rabbit polyclonal and mouse monoclonal anti-*PfAMA1* antibodies tested were consistently more inhibitory in the absence of *PfMSP2*. Of the two *PfAMA1* targeting i-bodies tested (Angage et al., 2024 [↗](#)), the most potent WD34 was also more potent in the absence of *PfMSP2*, with this activity increasing significantly with the ~5.6-fold increase in size that occurred upon addition of a human Fc region. Similarly, activity of a nanobody targeting the PCR complex protein CSS showed increased potency in the absence of *PfMSP2*, and again this was increased further with antibody enlargement through addition of an Fc region. These data suggest that *PfMSP2* may act to dampen the inhibitory activity of antibodies targeting other antigens on the surface of the merozoite. Such immune dampening (called conformational masking) by intrinsically disordered proteins, or protein regions, of host-cell invasion related proteins has recently been demonstrated by modelling and gene-editing studies in the viral pathogens HIV and HCV, respectively (Li et al., 2024 [↗](#); Stejskal et al., 2022 [↗](#)). These studies, along with our demonstration of increase antibody inhibitory efficacy with loss of the intrinsically disordered *PfMSP2*, support that this may be a widespread approach used by pathogens to protect important protein functional sites from inhibitory antibodies.

A role in immune evasion has previously been suggested for *PfMSP2* and speculatively an alternate explanation for the sensitisation observed is that this protein may shield key *PfAMA1* epitopes from antibody inhibition. For the *P. falciparum* 3D7 line, we found that *PfMSP2* knock-out led to higher *PfAMA1* binding as measured by increased fluorescence intensity for the mCherry tagged i-body WD34 in the absence of *PfMSP2*. Using a solid-phase model and titrating recombinant *PfMSP2*, we found that increasing concentrations of *PfMSP2* reduced binding of an anti-*PfAMA1* mAb and i-body, even at concentrations where *PfMSP2* was not detectable itself by antibodies. These data support that the presence of *PfMSP2* may reduce antibody access (masking) to important invasion proteins exposed on the merozoites surface. However, we acknowledge that the antibody-binding assays used (immunofluorescence assays, SPR, ELISA) might not be sensitive enough to observe all potential changes in binding, temporal changes when binding epitopes are exposed or differences in avidity that could lead to increased antibody potency during the course of parasite invasion and growth, and these factors may also contribute. Future studies of the merozoite surface and interactions between proteins may help decipher in full how *PfMSP2* is protecting essential merozoite antigens from invasion inhibitory antibody activity and reveal strategies for enhanced vaccine design.

Our observation that loss of *PfMSP2* potentiates the invasion-inhibitory activity of antibodies targeting other antigens opens an additional avenue for vaccine development, such as targeting *PfMSP2* to interfere with its potential role in immune evasion and thus potentiating antibodies against other merozoite antigens. Additionally, our results suggest that vaccines based on *PfAMA1* may need to be designed to specifically avoid the effect of *PfMSP2* modulating inhibitory antibodies, such as focussing on specific epitopes or structural features. To date, vaccines based on

AMA1 have failed to achieve substantial efficacy (Sagara et al., 2009 [↗](#); Sheehy et al., 2012 [↗](#); Thera et al., 2011 [↗](#)). We can speculate that one possible protective mechanism of action of the Combination B vaccine (Genton et al., 2002 [↗](#); McCarthy et al., 2011 [↗](#)), which combined *PfMSP2*, *PfMSP1* and *PfRESA*, could be that the inhibitory activity of vaccine-induced antibodies raised against *PfMSP2* were then able to potentiate the activity of antibodies targeting other antigens, potentially *PfMSP1* which was also a target of the vaccine. Here we show consistent potency improvement with *PfMSP2* knock-out for growth inhibitory rabbit, mouse monoclonal and i-body antibodies targeting *PfAMA1*, as well as demonstrate improved activity for and Fc-tagged nanobody targeting *PfCSS*, indicating that these are not outlier results from a single antibody or antibody type. However, increased antibody potency was not shared across all antibodies tested, possibly because the specific function or localisation of a target protein, the region that an antibody binds to or the functional activity (or lack thereof) of an antibody may all play a role in determining whether loss of *PfMSP2* can potentiate growth inhibitory activity. Further investigation using the parasite lines developed in this study and a wider panel of antibodies that target different stages of the merozoite invasion process could shed more light on this potentially novel mechanism of vaccine derived antibody efficacy.

The results of this study and work done by Escalante et al. (2022) [↗](#) demonstrate that MSP2-like sequences are found in the avian malarias *P. galinaceum* and *P. relictum*, which differs from the fields previous understanding that *PfMSP2* arose in the *Laverania* lineage of malaria parasites. This observation indicates that MSP2 was likely present in the ancestral malaria parasite before being lost in primate and rodent *Plasmodium* spp. Further insights into the evolution and loss of *PfMSP2* are likely to be found with additional whole genome sequences of bird and lizard malaria species. The presence of *msp2*-like sequences in some avian *Plasmodium* species and *Laverania* suggest a retention of *msp2* sequences in the *Laverania* which indicates an important role for MSP2 for these parasites. Indeed, our amino acid and structural prediction level comparison demonstrates that much of the protein properties conserved in the *Laverania* MSP2s are also recognisable, and in many cases conserved, in the avian malaria MSP2s, despite 10 million years of evolutionary divergence. This suggests that having an MSP2-like protein is beneficial for these parasites with very different hosts. However, CRISPR-Cas9 gene editing used in this work has shown that, in contrast to previous work, *PfMSP2* is not essential for *P. falciparum* blood stage parasite growth *in vitro*. Our findings raise the possibility that the retention of *PfMSP2* in the *Laverania*, and by association potentially the avian malaria parasites, may be linked to its apparent capacity to modulate the impact of antibodies against other merozoite surface-exposed antigens, rather than through a direct mechanistic role in host-cell invasion. If MSP2 was to fulfill such a role in the *Laverania* and avian malaria parasites, it raises the question of whether a similar protective system exists in other *Plasmodium* spp. and what the key protein/s are that fulfil this role in these parasites.

The *Pf3D7* MSP2 and *PfDd2* MSP2 lines developed in this study will provide useful reagents for investigating the potential of targeting a single, or both, *PfMSP2* allelic types with vaccines that elicit different immune responses. These parasites could also be used to explore whether vaccine-induced antibodies against other merozoite antigens could also be potentiated if any immune evasion mechanism provided by *PfMSP2* is removed. This may be achievable with a combination vaccine or through targeting an epitope of another merozoite vaccine candidate that is not protected by *PfMSP2*. Such information, investigated using the *PfMSP2* knock-out parasites developed in this study, could be useful in prioritising vaccine candidates, specific epitopes and combinations that target the merozoite.

Conclusion

Advancements in gene-editing techniques in *P. falciparum* have allowed us to show for the first time, and contrary to previous work, that *PfMSP2* is not essential for *P. falciparum* growth *in vitro*. Instead, we present a new concept that MSP2 can modulate the activity of invasion-inhibitory antibodies and may have evolved for this purpose. It is possible that other merozoite surface proteins may have similar roles and should be investigated in future studies. Our observation that

loss of *PfMSP2* potentiates the inhibitory activity of antibodies targeting other merozoite surface exposed proteins reveals a new avenue to explore in vaccine development by blocking or bypassing this function to improve the activity of vaccines targeting these antigens.

Materials and Methods

Bioinformatic analysis of *PfMSP2*, *PfMSP4* and *PfMSP5* like proteins in *Plasmodium* spp

The PlasmoDB database (Aurrecoechea et al., 2009 [link](#)) was used to identify in the available *Plasmodium* genomes the region between Adenylosuccinate lyase (PF3D7_0206700) and a conserved protein of unknown function (PF3D7_0207100) where *PfMSP2*, *PfMSP4* and *PfMSP5* are localised in *P. falciparum*. All annotated MSP2, MSP4 and MSP5 protein sequences were downloaded. Translated sequences of unannotated genes and putative annotations in this region were also downloaded and aligned using the MUSCLE alignment tool MEGA11: Molecular Evolutionary Genetics Analysis version 11 (Tamura et al., 2021 [link](#)) or the Geneious Prime 2019.2.1 MUSCLE alignment tool (Biomatters Ltd.) and determined to be MSP2, MSP4 or MSP5 based on sequence similarity in the conserved regions of these genes. A BLAST (NCBI) search with the N-terminal and C-terminal conserved regions of MSP2 was also performed to capture any sequences missed by the above approach. A maximum likelihood phylogenetic tree of identified MSP2 sequences was generated using the default maximum likelihood settings of MEGA11 with 500 bootstraps and the *P. gallinaceum* MSP2 as an outgroup.

Modelling of MSP structure across *Plasmodium* spp. using Alpha-fold

In order to obtain models of the 3-dimensional protein structures of MSP2 for *P. falciparum*, *P. reichenowi*, *P. billcollinsi*, *P. adleri*, *P. relictum* and *P. gallinaceum*, we first removed the predicted N-terminal signal sequences and C-terminal GPI anchor sequence. The resulting sequences were submitted to AlphaFold2 (Jumper et al., 2021 [link](#)). The resulting models were subsequently analysed and visualised using PyMOL 2.4.0 (Schrödinger, USA).

Culture and synchronisation of *P. falciparum*

Plasmodium falciparum 3D7 and Dd2 was cultured in RPMI-HEPES media (Gibco) containing 0.5% Albumax II (Gibco), 25 µM Gentamicin (Gibco), 367 µM hypoxanthine (Sigma Aldrich), 2 mM L-Glutamax (Gibco), 0.17% sodium bicarbonate (Sigma Aldrich) and O positive human red blood cells (Lifeblood, Australia) at 3% parasitemia, 3% haematocrit. Cultures were grown at 37°C under 1% oxygen, 5% CO₂ (BOC gas), in a humidified chamber. For gene-edited lines, WR99210 (Jakobus Pharmaceuticals) was added to culture media. Sorbitol lysis (5%) (Sigma Aldrich) to select for ring-stage parasites, Percoll Plus (Sigma Aldrich) selection of late-stage parasites and heparin treatment (Pfizer) (Boyle et al., 2010b [link](#)) were used to maintain synchronicity of parasite cultures.

Generation of *PfMSP2* knockout line using CRISPR Cas9

Briefly, a flank region incorporating the 5' region upstream of *PfMSP2*, the first 44 amino acids of the protein, and a flank of the region immediately 3' of *PfMSP2* was inserted into the pCC1 plasmid by restriction enzyme cloning. A 20 bp guide sequence was designed using EuPaGDT (Peng and Tarleton, 2015 [link](#)), annealed and inserted in the pUF_Cas9 plasmid by infusion cloning (Takara Bio). pUF_Cas9 guide plasmid (20 µg) and pCC1 *PfMSP2* repair plasmid (60 µg) was transfected directly into *P. falciparum* schizonts. Early schizonts were purified using a 70% percoll gradient and treated for 4 hrs with compound 1 (Taylor et al., 2010 [link](#)) at 2 µM. After incubation treated schizonts were washed twice and left shaking for 20 minutes before pelleting and resuspension in Complete Cytomix (plasmids plus 0.895% KCl, 0.0017% CaCl₂, 0.076% EGTA, 0.102% MgCl₂, 0.0871% K₂HPO₄, 0.068% KH₂PO₄, 0.708% HEPES). Schizonts were electroporated in 0.2 cm cuvettes (Biorad) at 800 V and 25 µF. Electroporated schizonts were mixed with warmed media and fresh

RBCs and shaken to promote invasion for 20 minutes after which they were placed in a culture dish. Transfected parasites were then treated with WR99210 to select for integration of the hDHFR drug selection cassette and MSP2 knock-out.

To confirm integration gDNA was collected by saponin lysis of schizont stage parasite culture and gDNA extracted using the PureLink Genomic DNA Mini Kit (Invitrogen) as per manufacturer's instructions. Integration of the *PfMSP2* knockout construct into a portion of parasites was confirmed by PCR (Supplementary Table 1 [↗](#)). Once confirmed cultures were cycled on and off WR99210 (Jakobus Pharmaceuticals) and 1 μ M 5FC (Sigma Aldrich) to select for integrated parasites no longer expressing the guide plasmid. Limited dilution cloning was then performed to obtain parasite clones which had integrated the knockout construct.

Western blotting to detect *PfMSP2*

Synchronised schizonts were harvested and the RBCs lysed by saponin. Briefly ~10 mL schizont (38-44 hrs) culture was lysed on ice for 10 minutes with 0.15% w/v saponin before pelleting by centrifugation and washing once in 0.075% w/v saponin followed by three washes in PBS containing protease inhibitors (CØmplete, Roche). Saponin treated schizont pellets were treated with DNaseI (Qiagen) for 10 minutes at room temperature before resuspension in reducing sample buffer (0.125 M Tris-HCL, 4% SDS, 20% glycerol, 10% beta-mercaptoethanol and 0.002% bromophenol blue). Proteins were separated by size using SDS-PAGE 4-12% Bis-Tris Gels (Bolt, Invitrogen) at 110V for 80 minutes before transfer to a nitrocellulose membrane (iBlot, Invitrogen) at 20V for 7 minutes. Blots were blocked from 1 hr to overnight in 3% skim milk PBS (Sigma Aldrich) before incubation with primary antibodies (1/75000 mouse 2F2 anti-*PfMSP2* 3D7 (Supplementary Table 2 [↗](#); (Adda et al., 2012 [↗](#))) and 1/75000 rabbit anti-*Plasmodium* aldolase (Abcam) or 1/75000 rabbit anti-FC27 and 1/10000 mouse anti- EXP2 (a gift of Paul Gilson, Burnet Institute, Melbourne) for 1- 2 hrs. IRDye 800CW goat anti-mouse (1/4000, LI-COR Biosciences) or IRDye 680RD goat anti-rabbit (1/4000, LI-COR Biosciences) secondary antibodies were used for detection on the Odyssey Infrared imaging system (LI-COR Biosciences). Quantification was performed using Image Studio Lite 5.2.5 (LI-COR Biosciences).

Quantification of parasite expansion rate and invasion inhibition

Potential growth defects resulting from gene editing was assessed by flow cytometry. The initial parasitemia of cultures was determined by flow cytometry and then measured again after the 50 mL cultures in 96 well plates were maintained under standard (still) or shaking (50 rpm) conditions for 48 hrs or 96 hrs of growth. To determine parasitemia by flow cytometry Ethidium Bromide (10 μ g/mL BioRad) was added to 50 μ L of 1% haematocrit culture for 30 mins to allow for staining of parasite DNA. The stain was washed off and wells resuspended in 200 μ L of PBS. Data was collected on a BD Accuri C6 Plus Flow Cytometer (BD Biosciences) and analysed on FlowJo (FlowJo LLC). Briefly, forward scatter and side scatter was used to determine the RBC population. From this population infected RBCs were gated on as highly fluorescent in the PE channel. Final parasitemia was compared back to initial parasitemia to determine the fold increase in parasitemia for each line.

To examine the invasion inhibitory effect of antibodies (Supplementary Table 2 [↗](#)), 5 μ L of potential inhibitor was mixed with 45 μ L of 0.2% schizont parasitemia at 1% haematocrit for 96 hrs before end parasitemia was determined by flow cytometry. Data is displayed as % Growth of media only controls for each line.

To examine differences in RBC receptor preference RBCs were treated with different enzymes that cleave different residues of RBC receptors which is used as a marker for which RBC receptor/s are preferred by *P. falciparum* (Duraisingh et al., 2003 [↗](#)). Infected RBCs at 1% ring stage parasitemia were washed three times in RPMI-HEPES to remove excess protein. Packed RBCs (20 μ L) at 1% parasitemia were then resuspended with 5X RBC volume of pre-warmed enzymes at desired concentration (Supplementary Table 3 [↗](#)). RBC-enzyme mix was incubated at 37°C for 45 mins with rotation to keep RBCs resuspended. Following incubation, RBCs were washed three times and added to fresh complete media at 1% haematocrit in a U- bottom 96 well plate in technical

duplicate. Plates were incubated for 72 hrs, sufficient time for *P. falciparum* to complete one full cycle of growth and reach trophozoite stage again. Final parasitemia was determined by flow cytometry, as detailed above, and compared to non-treated controls. IC₅₀ calculations data was log transformed then a nonlinear regression log-(inhibitor)-versus-response curve was calculated with Extra Sum-of-Squares F Test (best-fit LogIC₅₀) used to compared IC₅₀s between conditions.

Immunofluorescence assay of Wildtype and *PfMSP2* KO parasites

Schizonts ~38 hrs old were treated with E64 (Sigma-Aldrich) for 5 hrs to allow development of very mature schizonts desired for imaging. After E64 treatment, parasite cultures were fixed in 4% v/v paraformaldehyde (PFA, Sigma-Aldrich), 0.0075% v/v glutaraldehyde (pH 7.5, Electron Microscopy Sciences) solution for 30 min at room temperature shaking gently. Fixed parasites were washed in 1X PBS and then resuspended in PBS at 1% haematocrit. Coverslips (#1.5H high-precision coverslips, Carl Zeiss, Oberkochen, Germany) were coated in 0.01% Poly-L-lysine (Sigma Aldrich) for 1 hr at room temperature, washed in miliQ water before the fixed parasite culture was allowed to adhere for 1 hr. Cells were permeabilised by 0.1% Triton X-100 PBS for 10 mins before blocking for 1 hr to overnight in 3% BSA 0.05% Tween 20 PBS. Primary antibodies were diluted in 1% BSA 0.05% Tween 20 PBS and incubated for 2 hrs. Coverslips were washed in 0.05% Tween 20 PBS three times before incubation with the secondary antibody (1/500 goat anti-chicken/mouse/rabbit Alexa Fluor coupled secondary antibodies -488 nm, 594 nm, 647 nm; Life Technologies) for 1 hr. After the secondary antibody was washed off coverslips underwent a secondary fix in 4% v/v PFA, 0.0075% v/v glutaraldehyde for 5 mins. The fixative was washed off and the coverslips dehydrated with sequential 3 min ethanol treatment (70%, 90% and 100%). Coverslips were allowed to dry and mounted with ProLong Gold antifade solution (refractive index 1.4) which contains 4', 6-diamidino-2phenylindole, dihydrochloride (DAPI) (ThermoFisher Scientific) and allowed to set overnight. Imaging was performed on the Zeiss LSM 800 using the Airyscan super-resolution mode (Carl Zeiss, Oberkochen, Germany).

Live cell microscopic analysis of merozoite RBC invasion

Highly synchronised *PfDd2* and *PfDd2* ΔMSP2 parasites at schizont stages were adjusted to 0.25% Haematocrit in complete culture medium. A volume of 200 μL was added to a well of an iBidi 15 μ-Slide 8 well, glass bottom slide (iBidi 80827) and the slide was promptly returned to sit on a prewarmed water block within a gassed box and left to incubate at 37°C to allow iRBCs to settle. The slide was then transported directly to a prewarmed Nikon TiE microscope with an environmental chamber heated to 37°C and gas mixture of 1% O₂, 5% CO₂ and 94% nitrogen. Filming was conducted using either a 60X water or 100X oil objective. Differential interference contrast (DIC) imaging was captured at 3 – 3.5 Volts with a camera exposure of 60 – 180 milliseconds.

Captured footage was processed using the Nikon analysis software (NIS-Elements AR Analysis version 5.21.01) and analysis of the schizont rupture and merozoite invasion events was performed manually. The merozoites which invaded successfully were tracked and attachment time, reorientation time (if obvious), deformation start and end times, the deformation score (Weiss et al., 2015; Wilson et al., 2015), invasion initiation and completion times and echinocytosis start times were recorded.

RNA extraction

Highly synchronised cultures were pelleted, resuspended in TriZol (Invitrogen) and incubated for 5 minutes at 37°C. Chloroform (Sigma Aldrich) was added, mixed and then spun at 12,000g for 30 minutes at 4°C. Supernatant was collected and mixed with an equal volume of 70% Ethanol and processed using the RNeasy Kit (Qiagen) to extract purified RNA. RNA was DNase I (Qiagen) treated for 30 mins at room temperature and cleaned up on RNeasy columns according to the manufacturers protocol. Absence of gDNA was checked by qPCR of gDNA with primers (*PfSUB1*) and DNase I treatment repeated if necessary.

qPCR of gene expression in schizonts

DNase treated RNA was added to 40 mM dNTPs (Qiagen), 0.4 mg/mL random hexamer (Qiagen) and incubated at 65°C for 5 mins. After incubation 5x Superscript Buffer (Invitrogen), 100 mM DTT (Invitrogen), 40 U/ μ L RNaseOUT (Invitrogen) and 200 μ /mL Superscript III Reverse Transcriptase (Invitrogen) was added and cycled 25°C 5 mins, 50°C for 60 mins and 70°C for 15 mins. Primers (Supplementary Table 4 [\[4\]](#)) were designed to detect *Pf*MSP2, *Pf*MSP4, *Pf*MSP5, *Pf*SUB1 and Fructose Biphosphate Aldolase as a reference gene (Salanti et al., 2003 [\[5\]](#)). qPCR was performed on cDNA from WT and knockout parasites with PowerUp SYBR Green Master Mix (Applied Biosystems) and cycling parameters; 95°C for 10 mins followed by 40 cycles 95°C for 15 mins, 60°C for 14 mins before a dissociation step at 95°C for 2 mins followed by 60°C for 2 mins with a 2% ramp to 95°C for 2 mins on the QuantStudio 7 Flex System (ThermoFisher). qPCR was performed in triplicate for each of the three independent experiments. Change in gene expression in knockout lines compared to WT was quantified by the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen, 2001 [\[6\]](#)).

Differential gene expression in *Pf*MSP2 KO compared to WT schizonts

RNA sequencing of the schizont stage transcriptome of *Pf*3D7 MSP2 KO and *Pf*3D7 WT was performed using the Illumina total RNA with Ribo Zero plus for library preparation and sequenced on the NovaSeq 6000 platform (Victorian Clinical Genetics Services) with 2x150 bp paired end reads (Supplementary Table 5 [\[7\]](#)). Reads were aligned to the *Pf*3D7 reference genome (PlasmoDB v61) using STAR aligner (Dobin et al., 2013 [\[8\]](#)). Reads were called using Rsubread FeatureCounts (Liao et al., 2014 [\[9\]](#)) with reads counted by CDS and summarised by gene. Differential gene analysis was performed with DESeq2 (Love et al., 2014 [\[10\]](#)) and visualized in R studio using ggplots2 (Wickham, 2016 [\[11\]](#)). MSP2 RNASeq data is available at ArrayExpress through accession number E-MTAB-15427.

Quantitative Immunofluorescence assay

Synchronous *Pf*3D7 MSP2 KO and *Pf*3D7 WT parasite cultures were treated with ML10 (Ressurreição et al., 2020 [\[12\]](#)) to allow mature schizonts to develop. Thin blood smears were then made and fixed in 100%, -20°C methanol for 5 mins. Blocking of smears was performed in 1% BSA PBS, then fluorophore conjugated WD34 (240 ng/mL) and WD33 (120 ng/mL) i-bodies (Angage et al., 2024 [\[13\]](#)) in 1% BSA PBS were added and incubated at room temperature for 1 hr. Smears were then washed, ethanol dehydrated and a coverslip mounted (ProLong Diamond Antifade mountant with DAPI (refractive index 1.47, ThermoFisher Scientific)) and allowed to cure for 48 hrs at room temperature. Fluorescence images were acquired using an Olympus FV3000 confocal microscope equipped with a 100x oil objective (NA 1.4). Quantification of parasite associated fluorescence was performed using a modified image analysis pipeline (Liffner et al., 2020 [\[14\]](#)) implemented in ImarisTM (9.9 version) where non-specific background fluorescence was removed using rolling ball background subtraction and a thresholding algorithm applied to the image to generate binary masks, separating objects exceeding the threshold (putative parasites) from the background. Binary masks were further refined to exclude objects smaller or larger than a defined size range (< 3 μ m and > 15 μ m), corresponding to the expected size of AMA i-body stained schizonts. Following segmentation the surface area containing antibody signal were calculated and these values compared between *Pf*3D7 MSP2 KO and *Pf*3D7 WT parasites.

Surface Plasmon Resonance (SPR) Assay

Surface plasmon resonance (SPR) was employed to determine the association rate constant (K_a) of AMA1-antibody interactions in the presence of MSP2 using the Biacore T200 system (Cytiva). Flow cells 1, 2, and 3 were activated for 14 minutes using a 1:1 mixture of 0.1 M N-hydroxysuccinimide (NHS) and 0.4 M N-(3-dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (EDC) at a flow rate of 5 μ L/min. Flow cell one was immobilised with bovine serum albumin (BSA) to serve as a reference surface. AMA1 was immobilised at a concentration of 80 μ g/mL in 10 mM sodium

acetate (pH 4.5) to achieve a surface density of 1000 response units (RU) on flow cells Two and three. MSP2 was co-immobilised in flow cell three to a surface density of 2000 RU. All surfaces were subsequently blocked with a 7-minute injection of 1 M ethanolamine (pH 8.0). Purified antibodies were prepared in PBS containing 0.005% Tween-20 (pH 7.4) and injected over the flow cells at a flow rate of 60 $\mu\text{L}/\text{min}$ at 25°C. Single-cycle kinetics, using a bivalent analyte model, was applied to analyse the interactions.

Enzyme-Linked Immunosorbent Assay (ELISA)

ELISAs to evaluate the effect of MSP2 on the binding of AMA1-specific antibodies were carried out in a volume of 100 μL per well, either at room temperature for 1 hr or overnight at 4°C. All washing steps consisted of three washes with PBS containing 0.1% Tween-20 (PBS-T). Nunc Maxisorp plates (Thermo Fisher Scientific, USA) were coated with AMA1 at a concentration of 0.8 $\mu\text{g}/\text{mL}$, followed by washing and subsequent coating with increasing concentrations of recombinant HIS-tagged *Pf*MSP2, HIS-tagged *Pf*MSP4 or HIS-tagged NCAM. Wells were then blocked with 5% (w/v) skim milk in PBS and then incubated for 1 hr at room temperature with one of mouse Fc-conjugated WD34 anti-AMA1 i-body (Angage et al., 2024 [↗](#)), mouse monoclonal anti-AMA1 antibody (mAb) 4G2 (Kocken et al., 1998 [↗](#)), anti-MSP2 mAb 9G8 (Adda et al., 2012 [↗](#)), anti-HIS antibody (Sigma Aldrich) to detect *Pf*MSP4 or anti-NCAM antibody (21H5, (Griffiths et al., 2016 [↗](#))) at a concentration of 2.5 $\mu\text{g}/\text{mL}$. The primary antibody was washed off and the HRP-conjugated secondary antibody (1:5000, Sigma Aldrich) was then applied for 1 hr at room temperature before washing. For signal development to detect bound primary antibodies, wells were incubated with 1-StepTM Ultra TMB-ELISA Substrate Solution (Thermo Fisher Scientific, USA) according to the manufacturers instructions. The reaction was stopped by adding 1 M sulfuric acid, and the absorbance at 450 nm was measured using a microplate reader.

Statistical Analysis

Three independent experiments were performed for all experiments in technical duplicate, unless otherwise stated. Data was graphed and statistical tests performed in Prism GraphPad v9 or R.

Data availability

RNA sequence data were deposited in ArrayExpress (accession number E-MTAB-15427). Data generated or analysed during this study are included in the manuscript and supporting files and are available upon request from the corresponding author.

Acknowledgements

The authors would like to thank Prof Alan Cowman, Dr Stephen Scally and Jennifer Thompson (WEHI, Melbourne) for provision of PCR complex antibodies. Assoc Prof Paul Gilson (Burnet Institute, Melb) for EXP2 and MSP1 antibodies. Dr Chris MacRaild (Monash University) for helpful discussion. Dr Adam Thomas for the *Pf*MSP4 protein (Burnet Institute). Dr Sonja Frölich (the University of Adelaide) for assistance on quantitative immunofluorescence assays and Adelaide Microscopy for training and use of microscopy instruments. We thank Lifeblood, Australia, for provision of RBCs. DWW was supported by a Hospital Research Foundation (C-MCF-52-2019) and Australian Research Council (FT240100420) Future Fellowship. DWW and JGB by a Hospital Research Foundation Collaborative Grant (S-03-EOI-2021). JGB was supported by the National Health and Medical Research Council of Australia (Research Fellowship and Investigator Grant to JGB, Australian Centre for Research Excellence in Malaria Research). IGH and JC were supported by an Australian Research Council RTP scholarship. The Burnet Institute is supported by the NHMRC for Independent Research Institutes Infrastructure Support Scheme and the Victorian State Government Operational Infrastructure Support.

Additional information

Author Contributions

Conceived and designed the experiments: DWW, IGH, JC, JGB, DA, MF, RA. Performed the experiments and phylogenetic analysis: IGH, JC, KHL, OR, DA, KRT, NB. Analysed the data: IGH, DWW, JC, KHL, OR, DA, MF. Provided key discussions on PfMSP2 and/or key reagents: RA, MF, DA. Paper writing: DWW, JC, IGH, JGB with critical input from RA, MF, DA, OR, KRT and based on the PhD thesis of IGH.

Funding

Funder	Grant reference number	Author
Department of Education and Training Australian Research Council (ARC)	FT240100420	Danny W Wilson
Hospital Research Foundation (HRF)	C-MCF-52-2019	Danny W Wilson
Hospital Research Foundation (HRF)	S-03-EOI-2021	James Beeson Danny W Wilson
DHAC National Health and Medical Research Council (NHMRC)	2033320	James Beeson
DHAC National Health and Medical Research Council (NHMRC)	1134989	James Beeson
Department of Education and Training Australian Research Council (ARC)	RTP Scholarship	Isabelle G Henshall Jill Chmielewski

Author ORCID iDs

Danny W Wilson: <https://orcid.org/0000-0002-5073-1405>

Additional files

[Table 1](#) 

[Supplementary Figures 1 to 6](#) 

[Supplementary Tables 1 to 4](#) 

[Supplementary Table 5](#) 

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Peer reviews

Reviewer #2 (Public review):

The major strengths of the manuscript are in the *Plasmodium falciparum* genetic and phenotyping approaches. PfMSP2 knockouts are made in two different strains, which is important as it is known that invasion pathways can vary between strains, but is a level of comprehensiveness that is not always delivered in *P. falciparum* genetic studies. The knockout strains are characterised very thoroughly using multiple different assays and the authors should be commended for publishing a good deal of negative data, where no phenotype was detected. This is not always done but is very helpful for the field and reduces the potential for experimental redundancy, i.e. others repeating work that has already been performed but never published. The quality of the writing, referencing and figures is also generally strong.

There are certainly some areas of the manuscript that would benefit from deeper exploration, such as electron microscopy/other imaging approaches to explore whether deletion of PfMSP2 has a visible impact on merozoite surface structure, further replicates of the video microscopy assays to see whether trends in the data could reach significance (although these are very time-consuming and technically difficult assays), and follow up of some of the genes where expression is changed by PfMSP2 knockout (as the authors point out, there are no candidates that have a very obvious link to invasion suggesting that they may be compensating for PfMSP2 function, although several are expressed in schizont stages). However, there is already a substantial amount of data in the manuscript, and more detailed follow-up is reasonable to leave to future work. Overall, with the modifications made through the review process, including the addition of new controls for key experiments, the claims and conclusions are justified by the data, and the manuscript generates important new information about a highly studied *Plasmodium falciparum* merozoite surface protein.

<https://doi.org/10.7554/eLife.107603.2.sa2>

Reviewer #3 (Public review):

Henshall et al. study invasion of human erythrocytes by *Plasmodium falciparum* merozoites and report knockout of PfMSP2, a critical merozoite surface protein with unknown function. They describe conservation of MSP2 in *P. falciparum* and key avian malaria parasites, unabated growth of two knockout lines (Δ MSP2) produced in divergent 3D7 and Dd2 strains, no differences in expression of key invasion-associated genes, no effect on invasion kinetics (with or without protease treatment of erythrocytes), nonsignificant effects of knockout on parasite growth inhibition by antibodies directed against key invasion-associated antigens, and do find a significant effect on potentiating AMA1 invasion inhibitory antibodies. The studies are interesting and have potential for directing vaccine design targeting erythrocyte invasion, a critical step in bloodstream expansion of malaria parasites.

Major points:

(1) Much of the manuscript describes negative results and this reviewer found it arduous to get through many negative or nonsignificant results before finally getting to the significant effect on AMA1 inhibitory antibodies, not presented until Figure 6! Computational studies in Fig. 1 could be a supplementary figure. Figs. 2 and 3. demonstrate knockout in 3D7 and Dd2, respectively and could be assembled into a single figure. (Notably Fig. 2A and 3A are almost identical with use of some different primers.) Fig. 2E, 2F, 3D-H, all of Fig. 4, most of Fig. 5 are all negative or insignificant results that could also be moved to supplementary data. As MSP4, MSP5, and SUB1 are presumably included in the whole genome RNA-seq experiments shown

in Fig. 4C, it makes sense to remove Fig. 4A data from the paper fully. These consolidating changes would help highlight the key finding of improved binding and block of AMA1's role in invasion.

(2) The potentiating effects on anti-AMA1 antibodies are shown with rabbit sera and purified antibodies, mouse monoclonal antibodies, and smaller i-bodies inspired by shark antibody-like receptors but not with human monoclonal antibodies (hmAbs). As naturally acquired hmAbs targeting AMA1 have been identified and characterized (PMIDs: 39632799, 40020675), would it not be important to test these antibodies in the Δ MSP2, especially as the authors emphasize the importance of their model in designing better human malaria vaccines?

(3) Fig. 7 presents quantitative fluorescence microscopy to measure anti-AMA1 binding and support a model where MSP2 serves to sterically hinder antibody access to AMA1 on individual merozoites. I understand that the negative WD33 control is useful to contrast to the positive WD34 antibody (both bind AMA1 but only WD34 exhibits parasite growth inhibitory effects), but it seems that use of smaller i-bodies rather than conventional larger mouse or ideally human monoclonal antibodies may compromise demonstration of steric hindrance by MSP2 because smaller i-bodies may be less hinder.

(4) Some explanation for why WD33 fails to inhibit growth despite targeting the same antigen as WD34 is needed. Are the epitopes known? Does one bind further from the RON2 binding pocket?

<https://doi.org/10.7554/eLife.107603.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Henshall et al. delete the highly abundant merozoite surface protein PfMSP2 from two Plasmodium falciparum laboratory lines (3D7 and Dd2) using CRISPR-Cas9. Parasites lacking MSP2 replicate and invade red cells normally, opposing the experimental history that suggests MSP2 is essential. Unexpectedly, the knock-outs become more susceptible to several inhibitory antibodies - most strikingly those that target the apical antigen AMA1-while antibodies to other surface or secreted proteins are largely unaffected. Recombinant MSP2 added in vitro can dampen AMA1-antibody binding, supporting a "conformational masking" model. The reported data suggest that MSP2 helps shield key invasion ligands from host antibodies and may itself be a double-edged vaccine target.

Reviewer 1 did not have any comments we needed to address.

Reviewer #2 (Public review):

(1) The section describing Laverania and avian Plasmodium MSP2 comparison is a lengthy section and could be told much more concisely for clarity in delivering the key message, i.e., that conservation in distantly related Plasmodium species could indicate an important function. The identification of MSP2-like genes in avian Plasmodium species was highlighted previously in the referenced Escalante paper, so it is not entirely novel, although this paper goes into more detailed characterisation of the extent of conservation. Overall, this section takes up much more space in the manuscript than is merited by the novelty and significance of the findings.

As outlined in point (1) for Reviewer 1 (Recommendations for the authors), we have cut back through this section and focussed on the important comparisons rather than the general observation. We have also moved the elements of Table 1 to Supplementary Figures 2, 3 and 4 to streamline the manuscript. Further description of the changes is available in the Reviewer #1 (Recommendations for the authors).

(2) Characterisation of the knockout strains is generally thorough, though relatively few interactions were followed by live microscopy (Figures 3E-H). A minimum of 30 merozoites were followed in each assay (although the precise number is not specified in the figure or legend), but there are intriguing trends in the data that could potentially have become significant if n was increased.

In the Figure 3 Legend we have now indicated the number of merozoite invasions followed as per the following:

“(E-H) Key parameters of merozoite invasion were measured for both *PfDd2* WT (n = 43) and *PfDd2* Δ MSP2 (n = 35) parasites that had successfully invaded a RBC using live cell imaging of merozoite invasion.”

We have also removed the more general description of ‘a minimum of 30 merozoites’ from the same Figure Legend.

The number of schizont ruptures and subsequent merozoite invasions followed for each experiment is in line with previous studies that have investigated phenotypes with invasion inhibitors and gene knock-outs (e.g. Weiss et al. 2015, PLoS Pathogens). It is important to note that the data refers to merozoites that have completed invasion, and not just the number of merozoites that have been released from a schizont which is typically 2-4 times more than have invaded. This means we are comparing the kinetics of invasion across a relatively large sample size compared to other studies of inhibitory phenotypes. While it is possible that increasing the number of merozoites being filmed might lead to some statistical significance for some of the trends, we note that there is a limited growth phenotype overall in both short and long-term culture and this fits with the limited defect we are seeing. In order to better address this, as outlined in our response to point (7) for Reviewer 2 (Recommendations for the authors), we now discuss the trends seen in the data in additional detail.

*(3) The comparative RNAseq data is interesting, but is not followed up to any significant degree. Multiple transcripts are up-regulated in the absence of *PfMSP2*, but they are largely dismissed because they are genes of unknown function, not previously linked to invasion, or lack an obvious membrane anchor. Having gone to the lengths of exploring potentially compensatory changes in gene expression, it is disappointing not to validate or explore the hits that result.*

While we understand the reviewers comment, as outlined in the text we did not identify any upregulated proteins that looked like strong candidates to compensate for loss of MSP2 to explore in this manuscript. Instead, we chose to further investigate any potential loss of MSP2 phenotype that yielded the observations around improved potency of antibodies targeting some merozoite antigens with loss of MSP2. This will be explored in future studies as we try and understand the role of MSP2 in more detail and the interactions between proteins and antibodies on the merozoite surface.

*(4) Given the abundance of *PfMSP2* on the merozoite surface, it would have been interesting to see whether the knockout lines have any noticeable difference in surface composition, as viewed by electron microscopy, although, of course, this experiment relies on access to the appropriate facilities.*

We agree with the reviewer, but this lies outside the scope of this manuscript and optimisation of the imaging platform used to gain biologically useful insights would take a considerable amount of work based on feedback from people working with these techniques.

(5) One of the key findings is that deletion of PfMSP2 increases inhibition by some antibodies/nanobodies (some anti-CSS2, some anti-AMA1) but not others (anti-EBA/RH, anti-EBA175, anti-Rh5, anti-TRAMP, some anti-CSS2, some anti-AMA1). The data supporting these changes in inhibition are solid, but the selectivity of the effect (only a few antibodies, and generally those targeting later stages in invasion) is not really discussed in any detail. Do the authors have a hypothesis for this selectivity? The authors make attempts to explore the mechanisms for this antibody-masking (Figure 7), but the data is less solid. Surface Plasmon Resonance was non-conclusive, while an ELISA approach co-incubating MSP2 and anti-AMA1 antibodies to wells coated with AMA1 lacks appropriate controls (eg, including other merozoite proteins in similar experiments).

As outlined in our response to point (7) for Reviewer 2 (Recommendations for the authors), we have repeated the ELISA based assessment of recombinant MSP2s impact on anti-AMA1 antibody binding. In addition, we have included two comparator control proteins, the intrinsically disordered MSP4 of *P. falciparum* and the globular domain of the neural cell adhesion molecule (NCAM, CD56, 16 kDa), and found these proteins did not impact binding of anti-AMA1 antibodies. This strengthens the data that links the presence of MSP2 to reduced activity of anti-AMA1 antibodies.

As covered in our response to point (7) for Reviewer 2 (Recommendations for the authors) we provide additional discussion of this phenotype. We note that the list of inhibitory antibodies tested is not exhaustive, and additional antibodies may be identified where loss of MSP2 could improve potency. So although we see a consistent effect with a relatively small number of antibody targets, this does not rule out additional examples that may act earlier in invasion (for example, we noticed a small, but not statistically significant, trend for mildly inhibitory antibodies targeting MSP1-19 as well) and this makes speculating on why these two initial antibody targets at this time problematic.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) If feasible, perform ex vivo assays to demonstrate that the masking effect operates with physiologically relevant antibodies.

For this manuscript, we focussed on characterising the MSP2 knock-out parasites using the best reagents available. We remain interested in understanding whether these lines can be used to investigate the activity of functional antibodies from malaria exposed human serum and this will be the subject of future studies.

Reviewer #2 (Recommendations for the authors):

(1) As noted in the Public Review, the section describing MSP2 orthologues in other Laverania and avian Plasmodium species is overly long and not the most novel section of the manuscript. It could be really radically trimmed back.

We have taken this suggestion for the reviewer on board and have significantly cut back on our descriptions of the basic similarity properties of the conserved N and C-terminal regions as well as the description of the central variable region. Effectively, we have cut back the number of words through this section from 864 across 3 paragraphs to 478 across 2 paragraphs. While we have chosen to greatly economise our description of the N and C-terminal conserved regions, we have maintained much of the description of the similarities

and differences in the central variable region as we believe the observation that this variant region still maintaining repeats, though they differ in size, number and amino acid composition, across such evolutionary distances is of interest.

Taking the reviewers comment on board, we have also removed Table 1 from the manuscript (shows amino acid sequence properties of these regions) and instead have inserted the tables relevant for each alignment in Supplementary Figures 2, 3 and 4 as appropriate. This will streamline the main manuscript and better align amino acid property and alignment data in the one Figure. We thank the reviewer for this feedback and believe that this has helped focus the text on the most important observations.

(2) Figure 2C - As MSP2 has stage-specific expression, it could be informative to incorporate an antibody targeting another gene with a similar stage-specific expression pattern, such as AMA,1 into the blot. This would confirm that both protein samples were collected at a similar point during blood stage development.

We have modified Figure 2C to include both the original comparison using *PfAldolase* as the loading control and also the merozoite expressed *PfGAP45* as a loading/stage specific control as per the Figure.

*(3) Figure 2D - Magenta and red are hard to distinguish in the merge channel. Is it possible to pseudocolour one of these channels a different colour? Also, it would be simpler to keep *PfMSP2* a consistent colour in both rows.*

Thank you for this suggestion and we agree that the comparison could be made clearer. For this figure, we have coloured DAPI to label the nuclei (Cyan), and antibodies targeting *PfMSP2* (Magenta), *PfAMA1* and *PfMSP1-19* (Yellow). This is also reflected in the merged image. The Figure legend now reads:

“(D) Distribution of key merozoite surface proteins in the presence or absence of *PfMSP2* was visualised by immunofluorescence. *PfMSP2* (magenta), the nucleus stained by DAPI (cyan) and *PfAMA1* (yellow, top two rows) or *PfMSP1-19* (yellow, bottom two rows), and the coloured merge of the preceding panels. Scale bar = 0.7 μ m. Representative images shown from a minimum of 10 schizonts imaged per condition.”

(4) Figure 2F - Static growth relative to shaking growth is plotted in this panel; perhaps this could be more clearly described in the legend or mentioned in the text that there was not a significant alteration in growth in static or shaking conditions.

As suggested, we have clarified the result in the Figure legend text as follows:

“(E-F) Growth of *Pf3D7* WT compared to *Pf3D7* Δ MSP2 *P. falciparum* parasites, measured as fold increase in parasitaemia, over one (48 hrs) or two (96 hrs) cycles in either standard (still- (E)) or shaking (F) conditions, with no measurable difference between parasite growth rates seen between standard or shaking conditions.”

Please also describe the shaking conditions used (i.e., speed, culture size, and vessel) in the methods.

We have updated the methods to provide information on the growth conditions used in the standard versus shaking growth assays:

“The initial parasitemia of cultures was determined by flow cytometry and then measured again after the 50 mL cultures in 96 well plates were maintained under standard (still) or shaking (50 rpm) conditions for 48 hrs or 96 hrs of growth.”

(5) Figure 3G - Annotate legend for strength of deformation to describe what 1,2, or 3 refers to.

We have added the following to the Figure legend of Figure 3G:

“Deformation scores are as defined by Weiss et al (Weiss et al., 2015), with 1 = weak deformation of the RBC membrane at the point of contact, 2 = strong deformation leading to the RBC membrane extending up the sides of the merozoite and changes in RBC membrane curvature beyond the point of contact and 3 = extreme deformation indicated by the merozoite being deeply embedded in the RBC membrane and strong deformation of the RBC well beyond the point of contact.”

There is a small visible shift in the deformation event scores. Is this also not significant? Even if deformation is not significantly longer, could this small effect alter the exposure of epitopes on other proteins for antibody targeting?

We did test the deformation event scores and the differences were non-significant. We have considered this possibility raised by the reviewer, but we are cautious in over interpreting the possibility that these trends might contribute to the increased potency of certain antibodies in the absence of additional data. We note that, although deformation may happen over a slightly longer timescale and show more aggressive deformations with *PfMSP2* knock-out, this also seems to translate into a weak trend for faster overall entry for those merozoites that go on to invade. Therefore, although deformation may be longer and stronger, antibodies may have less time to block invasion overall. We are not confident that we can interpret around what might be happening at the molecular scale here based on this data and have chosen not to discuss this possibility in the manuscript. However, we have added the following to the results to better explain the phenotype the phenotype we observed.

“This analysis showed that, although there was a trend for *PfDd2* Δ MSP2 knock-out parasites to have a higher mean time to attach to the RBC, as well as for the length and strength of RBC deformation, these trends did not reach significance. For those merozoites that did invade the RBC, on average it took less time for *PfDd2* Δ MSP2 knock-out parasites to invade than *PfDd2* WT, but this again did not reach significance (Figure 3 E-H). Together these data show *PfMSP2* is not essential for blood-stage replication *in vitro* in two *P. falciparum* laboratory isolates from different geographical regions and knock-out of *PfMSP2* does not seem to significantly impact parasite growth or merozoite invasion *in vitro*.”

(6) Figure 4C - Legend refers to black lines, but on the figure, they are red? Is the horizontal red line in the correct place, or should some of the dots below it be black rather than blue if they fall outside the adjusted p-value significance cut-off? Were 4 schizont harvests performed in total, or 4 for each cell line?

We thank the reviewer for pointing this out and we have now changed the text to say red lines. We have also provided more information in the Figure legend to more clearly define what data is represented. In short, 4 harvests were performed for each cell line (8 in total across the 2 cell lines) and the data represents the distribution from one of these harvests. The blue shaded genes are those that, on average, across the 4 *Pf3D7* WT and *Pf3D7* Δ MSP2 paired harvests show up or down-regulated expression. This is why some of the blue shaded genes lie near or below the cut-off values represented by the red line. The Figure legend text has now been modified as follows.

“(C) Log2(fold change) for differentially expressed genes, including multigene families, between the transcriptome of *Pf3D7* WT and *Pf3D7* ΔMSP2 schizonts. Plot represents the results for one of four independent schizont RNA harvests for *Pf3D7* WT and *Pf3D7* ΔMSP2 parasites and red lines differentiate genes with a log2 (fold change) > 0.5 and < -0.5 with adjusted p-value < 0.01. Genes shaded blue represent those genes that were found to have an average log2 (fold change) > 0.5 (dark blue) or < -0.5 (light blue) across the four replicate samples compared. Significance determined as below $p < 0.05$ after correction for multiple testing.”

(7) Figure 7D - ELISA results don't show a convincing concentration-dependent inhibition, and repeating with another recombinant protein is essential before inferring that the effect is specific to PfMSP2

We have repeated the ELISA experiment using recombinant PfMSP2 to reduce variability across the assay and again found a dose dependent reduction of anti-PfAMA1 binding with increasing concentrations of recombinant PfMSP2. It should be noted that this is a completely new set of experiments that recapitulate the original findings. See updated Figure 7D.

We agree with the reviewer that the experiment and interpretation of the data would be strengthened by comparing any potential inhibitory impact on anti-PfAMA1 binding to a different recombinant protein. Therefore, we have completed identical experiments using the similarly intrinsically disordered PfMSP4 recombinant protein (40 kDa) and the highly structured 16 kDa immunoglobulin domain of human neural cell adhesion molecule (NCAM). We find that there is no dose dependent loss of anti-PfAMA1 binding to recombinant PfAMA1 with addition of PfMSP4 or NCAM immunoglobulin domain recombinant protein. These controls are contained in Supplementary Figure 6, the relevant text is provided below.

‘In contrast, increasing concentrations of the intrinsically disordered MSP4 from *P. falciparum* 3D7 (40 kDa) and the highly structured immunoglobulin domain of neural cell adhesion molecule (NCAM, CD56, 16 kDa) recombinant proteins did not impact on binding of anti-PfAMA1 antibodies to recombinant AMA1 (Supplementary Figure 6).’

(8) Again, as noted in the public review, the target-specificity of the inhibition-masking effect is perhaps the most surprising aspect of the data - this could do with much more thorough discussion. Why only these proteins, both of which function late in invasion?

Overall, we tested several growth inhibitory and non-inhibitory antibodies shown to bind specifically to individual or some combination of nine *P. falciparum* merozoite surface and secreted proteins. However, we do not consider this to be an exhaustive list of potentially invasion inhibitory antibodies by any means. We mostly did not observe any non-inhibitory antibodies becoming significantly more growth inhibitory to PfMSP2 KO lines, indicating that these antibodies were not impacted by loss of PfMSP2 or had no functional inhibitory effect in these assays.

What we do demonstrate here is that we see a consistent impact with different rabbit, mouse monoclonal and i-body growth inhibitory antibodies targeting PfAMA1, indicating that it is not a spurious result from a single antibody or antibody type. We also find a second example, with nanobodies targeting the PfPCRCR complex protein PfCSS potentiated with loss of PfMSP2. This opens up the possibility that other growth inhibitory antibodies to the antigens tested here, or growth inhibitory antibodies targeting other antigens involved in merozoite invasion, may also become more potent with MSP2KO. Although both PfAMA1 and PfCSS function late in invasion, it is too early to say whether this is a functional trend or an observation that is related to the panel of antibodies tested. Therefore, further testing using lines developed in this study could yield additional examples of antibodies that become more inhibitory with MSP2 KO and provide additional information on the potential impact that

MSP2 may have on their vaccine potential. In order to address this, we have added the following text to the discussion:

“Here we show consistent potency improvement with *Pf*MSP2 knock-out for growth inhibitory rabbit, mouse monoclonal and i-body antibodies targeting *Pf*AMA1, as well as demonstrate improved activity for and Fc-tagged nanobody targeting *Pf*CSS, indicating that these are not outlier results from a single antibody or antibody type. However, increased antibody potency was not shared across all antibodies tested, possibly because the specific function or localisation of a target protein, the region that an antibody binds to or the functional activity (or lack thereof) of an antibody may all play a role in determining whether loss of *Pf*MSP2 can potentiate growth inhibitory activity. Further investigation using the parasite lines developed in this study and a wider panel of antibodies that target different stages of the merozoite invasion process could shed more light on this potentially novel mechanism of vaccine derived antibody efficacy.”

(9) Typos/minor editorial points:

L111 – conserved

This text has been modified.

L235-237 - check the wording in this sentence for clarity

This text has been modified.

Figure 3E - 'attachment' on axis

This Figure has been modified.

L350 - mentions eight 'proteins' having expression increase, instead 'transcripts' should be referred to when describing RNAseq data, as transcript levels may not correspond directly with protein levels. Also, be careful when referring to transcript or protein throughout this paragraph.

This text has been modified.

Figure 4A - instead of 'transcription during schizonts', better to say 'schizont transcript abundance'

This text has been modified.

L514 - 'detectable binding to *Pf*AMA1'

This text has been modified.

L589 - Is it a mouse Fc region or a human Fc region that is added? The human Fc region is mentioned in the results.

In the growth inhibition assays anti-AMA1 WD34 i-body with a human FC region was used and in the ELISA assays anti-AMA1 WD34 i-body with a mouse FC region (to enable detection of AMA1 binding use the same secondary anti-body for both the WD34 i-body and the 4G2 mouse monoclonal antibody) was used. The text has been checked and modified accordingly to clearly say this.

Supplementary figure 3 - 'repeats'

This text has been modified.

<https://doi.org/10.7554/eLife.107603.2.sa0>